

Trajectory-Prediction Techniques for Unmanned Aerial Vehicles (UAVs): A Comprehensive Survey

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Abstract—Unmanned Aerial Vehicles (UAVs) have witnessed remarkable significance in diverse sectors, ranging from environmental monitoring, infrastructure inspection, disaster response, wildlife conservation, surveillance, and reconnaissance missions. It is crucial to predict their future states to enable UAVs' safe and efficient operation in dynamic environments. UAV trajectory planning is a crucial aspect of UAV operations, as it determines how the drone will navigate, perform tasks, and avoid obstacles. UAVs can be operated with varying degrees of autonomy, and they can be controlled by humans or autonomously via onboard autopilot software. While existing research has extensively focused on trajectory planning methodologies for UAVs, there is a noticeable gap in the literature concerning the integration of predictive capabilities into trajectory planning, highlighting the need for a comprehensive review of methodologies in UAV trajectory prediction connected with the associated realm of trajectory planning. This article provides a comprehensive and comparative analysis of trajectory prediction methods tailored for autonomous UAVs. Beginning with a precise problem definition and algorithm categorization, our study delves into evaluating methodologies rooted in conventional mathematical models, classical machine learning, deep learning, and reinforcement learning models.

Index Terms—UAVs, trajectory prediction, machine learning, deep learning, reinforcement learning, autonomous, dynamic environment, surveillance, optimization, security, reliability.

I. INTRODUCTION

UNMANNED Aerial Vehicles (UAVs) have attracted growing interest from both the academic and industrial sectors due to their capacity to tackle a range of enduring challenges in the field of transportation and communication [1], [2], such as safety [3], [4], [5], congestion [5], [6], [7], and energy efficiency [8], [9]. In recent times, substantial progress has been achieved in the advancement of perception, planning, and control systems [10] for UAVs. However, the widespread adoption of UAVs will only become a reality if their autonomous capabilities are thoroughly validated for safety [11], [12], [13], [14], [15], [16].

The challenges faced in UAV operations are multifaceted, incorporating complex traffic maneuvers [2], [17], intricate

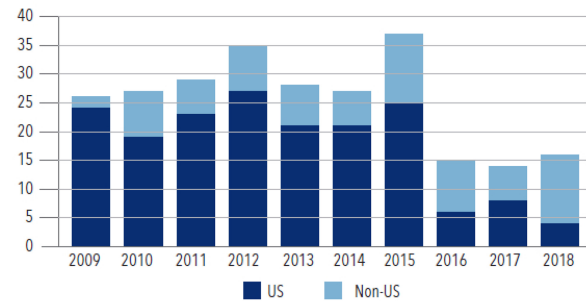


Fig. 1. Drone Wars UK Dataset: Number of drone crashes in U.S. vs Non-U.S. [25].

interactions [3], sensor uncertainties [18], [19], computational demands [13], [20], and tight timeframes [21], [22]. Neglecting these challenges can lead to severe accidents, endangering lives and property [23], [24]. A report by Drone Wars UK shows that a surge in large military drone accidents, with over 250 documented crashes as shown in Fig. 1, which adds valuable context and safety considerations for improving trajectory prediction models in both military and civilian UAV applications [25]. Collaborative efforts by researchers, policymakers, and industry are imperative to ensure UAV advancements with minimized accident risks. UAV trajectory prediction can help in tackling these challenges by providing real-time forecasts of the UAV's flight path and allowing for proactive decision-making to avoid potential hazards [26].

Trajectory prediction enables UAVs to anticipate future environmental states and plays a crucial role in enhancing safety in UAV operations [1], [27], [28], [29]. Much like human drivers who predict the movements of nearby vehicles before making driving decisions, UAVs must analyze and predict future trajectories [30]. These predicted trajectories serve as invaluable indicators for early hazard detection [31], influencing decision-making [32], [33] and planning algorithms [34], as shown in Fig. 2, along with the different ways of data collection.

Machine learning (ML) [35], [36], [37], Deep Learning (DL) [30], [38], and Reinforcement Learning (RL) [39], [40], [41] algorithms have significantly influenced trajectory prediction, especially in the analysis of UAV data. DL techniques like object detection and tracking in UAV environments have been applied in many studies. For Example, You Only Look Once (YOLO) and Single Shot MultiBox Detector (SSD)-based Single Object Detection (SOD) approaches have been applied in [42], [43], [44]

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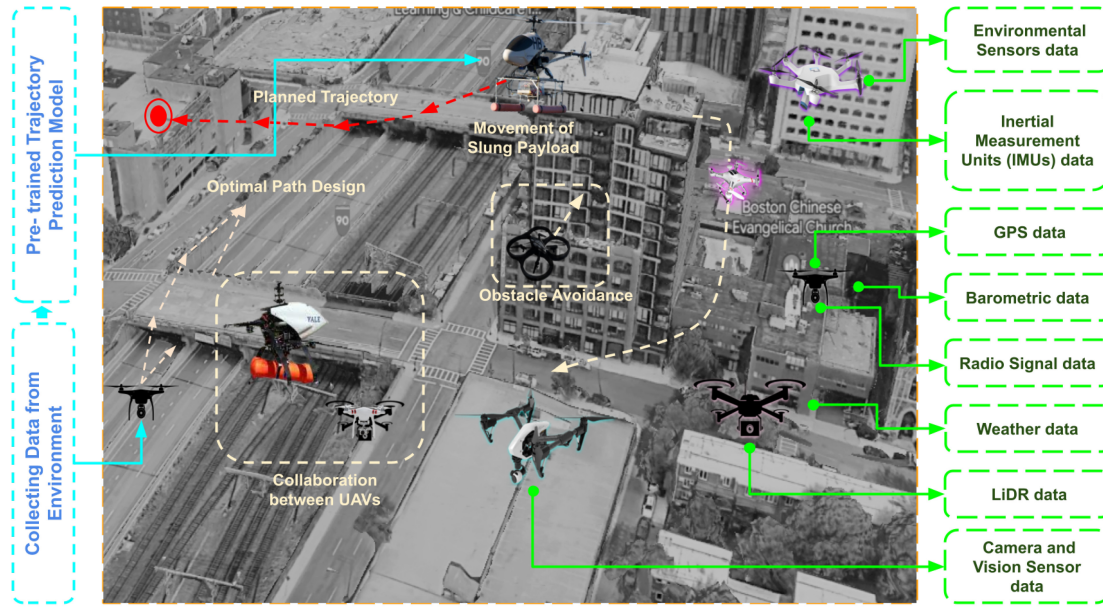


Fig. 2. A visual representation of Multi-UAV Environment, Data Collection, Challenges and Prediction. (Background: Boston City, U.S., online access on Google Earth).

due to their high efficiency. Researchers are employing diverse neural network architectures to enhance the accuracy of trajectory prediction and optimize efficiency [45], [46]. Numerous studies propose predictive models for optimizing UAV communication locations by considering wind [4], [47], time delays [4], [22], [48], antenna arrays [49], [50], [51], and Direction of Arrival (DOA) estimation algorithms [26], [52]. These strategies hold promise for a wide range of applications, spanning both military and civilian domains, including applications such as aerial mapping and environmental monitoring [46], [53]. Furthermore, by combining error budgeting and mitigation strategies, various researchers try to improve Trajectory Prediction (TP) models for UAVs [54], [55], [56].

The applications of trajectory predictions also involve handling uncertainty quantification and rapid changes in flight patterns [57]. In the realm of UAVs, there is a pressing need for further advancements in precise Video Object Detection (VID) [58] and Multiple Object Tracking (MOT) [59], [60] techniques explicitly tailored for UAV environments. The study [61] used trajectory planning in a dynamic environment of a UAV-enabled data collection model in Wireless Sensor Networks (WSN), overcoming the limitation of static WSN by replacing it with enabled WSN where trajectory planning plays an essential role in optimizing the maneuvering of UAVs to achieve efficient and effective data collection from WSNs.

In this introductory section, we comprehensively address the intricate challenges encountered during the UAV trajectory prediction process. Subsequently, we present our noteworthy contributions in this paper, encompassing a thorough examination of diverse trajectory prediction methodologies and their systematic categorization. To enhance the material flow, we present the paper's structure in Fig. 3, and the compilation of acronyms is available in Table I.

A. Challenges

This section offers an overview of the research's purpose and highlights critical issues in UAV trajectory prediction widely explored in existing literature. The challenges are:

1) *UAV Collision Avoidance*: Previous models often faced challenges ensuring safe UAV operations by avoiding collisions with other aircraft, obstacles, or neighboring UAVs, as mentioned by Hsu and Gau [62]. This study contributes to UAV trajectory prediction by introducing a distributed reinforcement learning approach to address collision avoidance challenges, reducing state-space complexity. The article [31] also talked about the security of aerial vehicles in future dense air traffic. The article [63] shows that the previous studies have limitations, such as focusing on specific localization methods, not considering varying importance levels, or lacking comprehensive joint trajectory and localization strategies. Previous studies have focused on static or known-user scenarios. However, the need for trajectory design based on users' trajectory prediction is emphasized for improved service quality and response time [64]. Many existing methods lacked consideration for obstacles, safety constraints, and global information, where some methods had high computational costs or were prone to local minima; other approaches suffered from overfitting, uncertainty, or inadequate motion models [65].

2) *Heterogeneous Ground IoT Devices and Data Collection*: Existing models assumed uniform communication radii for ground IoT devices, which may not align with real-world scenarios. Many researchers introduce the concept of ground IoT devices with varying communication radii, driven by data rate requirements [66]. This enhances the realism of IoT device distribution and presents data collection challenges more faithfully [66], [67] so that data can be collected accurately for trajectory prediction. The optimization

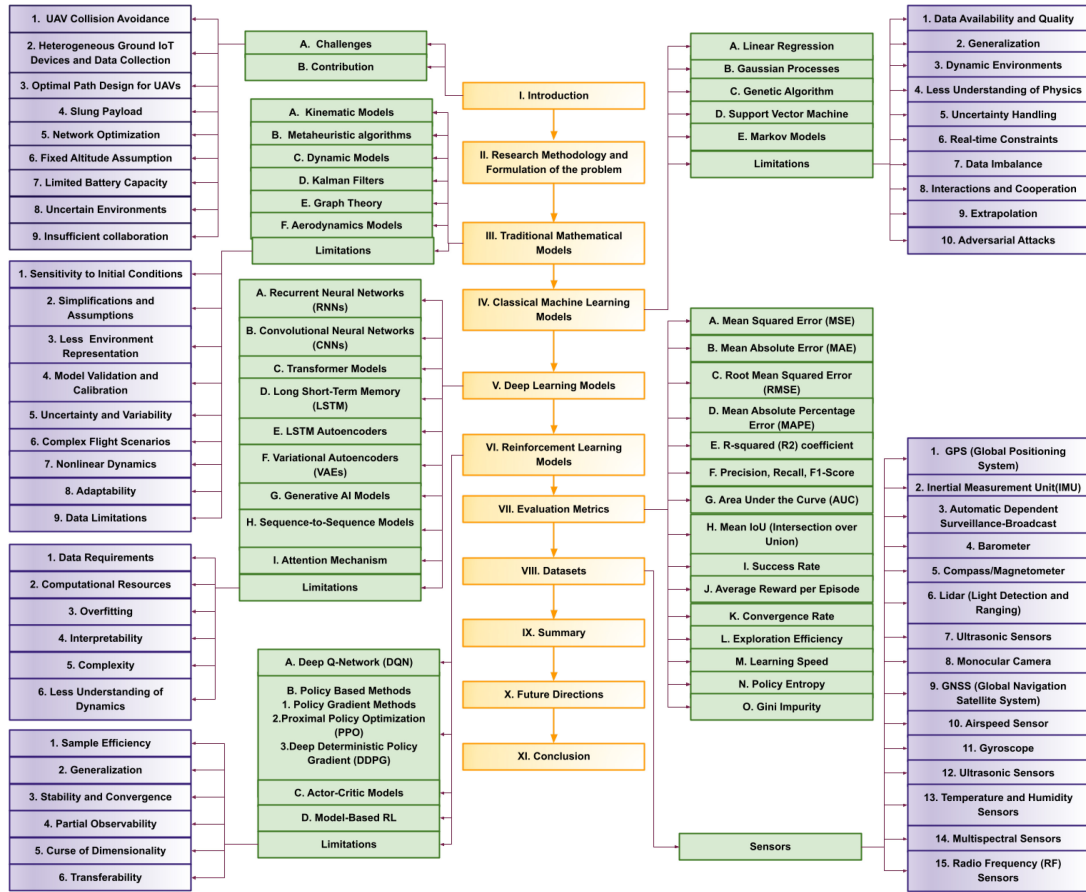


Fig. 3. Survey Structure and Topic Categorization.

problems for trajectory planning need to be extended to accommodate this heterogeneity. The accuracy of motion prediction for UAVs is limited due to the lack of training data, even though deep learning models are generally suitable for the task [27]. The study [21] aims to resolve the challenge of optimizing UAV trajectories for time-sensitive data collection from various heterogeneous IoT Devices, considering factors like maximum and average age of information. The study [68] highlights the limitations in existing methods, particularly those relying on Kalman Filters (KF) and traditional deep learning models like Long Short Term Memory (LSTM), for handling nonlinearities and noise in real-world scenarios. Previous models for drone camera localization faced challenges in processing time and computational demand [69].

3) *Optimal Path Design for UAVs*: Many works like [7], [62], [70], [71] addressed coverage or trajectory optimization separately, and some solutions were tailored to specific scenarios like static obstacles. The research [62] aimed to design optimal paths for UAVs that serve two purposes: transporting goods along the outbound route and gathering data from diverse ground-based IoT devices on the return journey. This dual-purpose optimization introduced novel challenges, including determining optimal visiting sequences and minimizing path lengths. Current Trajectory Prediction (TP) models that are rooted in motion equations are significantly dependent on pre-existing knowledge [72], leading

to limited accuracy in real-time flight scenarios. Data-driven TP models, which map inputs to outputs using machine learning, offer an alternative, but many are limited to one-step prediction. The primary challenge being addressed in [73] is the adaptability of Q values to account for changes in UAV behavior, such as sudden turns or maneuvers. Techniques like Sliding Mode Control (SMC) and adaptive control methods have attempted to address robustness and disturbance rejection. However, the intricate dynamics of drones in dynamic environments limit the applicability of these solutions [74]. The study [75] shows that many issues involve optimizing UAV altitude and location for QoS, along with shortening UAV-to-GU (Ground Unit) communication distances.

4) *Slung Payload*: Slung payload challenges in UAVs include aerodynamic disruption, swing, load distribution, payload safety, control algorithm adaptation, and operational limitations [76], [77], [78]. Overcoming these issues demands advanced control and stabilization measures for safe and efficient operations. The article [78] discusses the difficulty of controlling quadrotor UAVs with slung payloads due to their complex dynamics. The difficulty lies in achieving precise quadrotor positioning and simultaneous control of the payload swing.

5) *Network Optimization*: Existing works have explored UAV-assisted wireless communication for various applications, but challenges remain in optimizing UAV trajectories,

TABLE I
LIST OF ACRONYMS

Acronym	Definition	Acronym	Definition
2D	Two-Dimensional	LQG-MP	Linear Quadratic Gaussian Motion Planning
3D	Three-Dimensional	LSTM	Long Short-Term Memory
4D	Four-Dimensional	LSTM-AE	LSTM Autoencoders
ACD	Automatic Content Descriptor	MALS-Net	Multi-Head Attention-Based LSTM Seq2Seq Network
ACDQL	Actor-Critic Deep Q-Learning	MAE	Mean Absolute Error
ADS-B	Automatic Dependent Surveillance-Broadcast	MAJPPPO	MultiAgent Joint Proximal Policy Optimization
AG-PMADDPG	Air-Ground Probabilistic Multiagent Deep Deterministic Policy Gradient	MDP	Markov Decision Process
AIAA	American Institute of Aeronautics and Astronautics	MCS	Monte Carlo Simulations
AOA	Angle of Arrival	MDN	Mixture Density Network
AUC	Area Under the Curve	MPC	Model Predictive Control
ATS	Air Traffic Simulator	MPGA	Multi-Population Genetic Algorithm
ATSP	Asymmetric Traveling Salesman Problem	MTCDs	Machine Type Communication Devices
ATP	Aircraft Trajectory Prediction	MOT	Multiple Object Tracking
ATS	Air Traffic Simulator	MPC	Model Predictive Control
BCIA	Beijing Capital International Airport	MM	Markov Models
BRB	Branch, Reduce, and Bound	ML	Machine Learning
CA-E	Convolutional Autoencoders	MLP	Multilayer Perceptron
CAS	Collision Avoidance System	MLR	Multiple Linear Regression
C-AE	Convolutional Autoencoders	NOMA	Non-Orthogonal Multiple Access
CF-VAE	Conditional Flow Variational Autoencoders	NRI	Neural Relational Inference
CNN	Convolutional Neural Network	NN	Neural Network
CS	Cuckoo Search	OMA	Orthogonal Multiple Access
CS-CVAE	Cascaded Sliding Conditional Variational Autoencoder	PPO	Proximal Policy Optimization
CSIR	Cascaded Sliding CVAEs With Iterative Socially-Aware Rethinking	POA	Price of Anarchy
DCA	Dual Connectivity-Aware	PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
DDQN	Double Deep Q-Network	PRQoS	Predictive Quality of Service
DDPG	Deep Deterministic Policy Gradient	PSO	Particle Swarm Optimization
DDRRT	Domain Rapidly Exploring Random Tree	Q value	Quality Value
Deep ESN	Deep Echo State Network	QUAV	Quadrotor Unmanned Aerial Vehicle
DVAE	Descriptive Variational Autoencoder	QoS	Quality of Service
DNN	Deep Neural Network	R ²	R-squared
DL	Deep Learning	RAC	Recursive Area Clustering
DOA	Direction of Arrival	RF	Radio Frequency
DRQN	Deep Recurrent Q-Network	RNNs	Recurrent Neural Networks
DRL	Deep Reinforcement Learning	ROS	Robot Operating System
DQN	Deep Q-Network	RSSI	Received Signal Strength Indicator
DTTN	Deep Transformer Tracking Network	SAC	Soft Actor-Critic
EE	Exploration Efficiency	SCA	Successive Convex Approximation
ESB-MADDPG	Expert System-based Multiagent Deep Deterministic Policy Gradient	SD	ScienceDirect
ETA	Expected Time of Arrival	SMC	Sliding Mode Control
F450	Drone Model	SN	Sensor Node
FAA	Federal Aviation Administration	SNR	Signal-to-Noise Ratio
Fig.	Figure	SSD	Single Shot MultiBox Detector
GA	Genetic Algorithm	STOA	State of the Art
GAN	Generative Adversarial Network	TD3	Twin Delayed Deep Deterministic Policy Gradient
		TD3-TDCTM	TD3-based Trajectory Design for Completion Time Minimization
GDTs	Ground Data Terminals	TFMS	Traffic Flow Management System
GDAT	Graph Double-Attention Network	TMA	Terminal Maneuvering Area
GINI	Gini Impurity	TM	Transformer Model
GLONASS	Global Navigation Satellite System	TMN	Trajectory Mapping Network
GPR	Gaussian Process Regression	TP	Trajectory Prediction
GPS	Global Positioning System	TSCNN	Time-Series Convolutional Neural Network
GRU	Gated Recurrent Unit	TRPO	Trust Region Policy Optimization
GU	Ground Unit	UAS	Unmanned Aircraft Systems
H	Policy Entropy	UAV-BSs	UAV Base Stations
HHPS	Hybrid Hovering Positions Selection	UAVs	Unmanned Aerial Vehicles
HMM	Hidden Markov Model	UGV	Unmanned Ground Vehicle
I-SCR	Iterative Future-Past Social Compatible Rethinking	UK	United Kingdom
IEEE	Institute of Electrical and Electronics Engineers	US	United States
IG	Gini Impurity	V-REP	Virtual Robot Experimentation Platform
IMU	Inertial Measurement Unit	VAEs	Variational Autoencoders
IoU	Intersection over Union	VGG16	CNN model
IOT	Internet of Things	VID	Video Object Detection
LQ	Linear Quadratic	WSN	Wireless Sensor Network
		YOLO	You Only Look Once

resource allocation, and communication efficiency [79]. Many researches aimed to provide comprehensive solutions encompassing UAV trajectory planning, collision avoidance, and data collection from ground IoT devices, all while optimizing energy efficiency and minimizing completion times [35], [62], [79], [80], [81], [82], [83], [84], [85], [86]. The study [87] indicates that numerous models encountered difficulties in guaranteeing network connectivity and accurate data collection in situations involving the mobility of ground users or

the imprecise deployment of sensor nodes. The article [88] addresses that the critical challenge is to predict network topology changes due to the dynamic movements of UAVs and utilize this information for more efficient control and communication. Previous models lacked effective proactive and predictive routing mechanisms in dynamic UAV networks. Traditional approaches relied on current network topology, leading to frequent link failures and disruptions [87], [88]. UAV Base Stations (UAV-BSs) offer mobile and adaptable

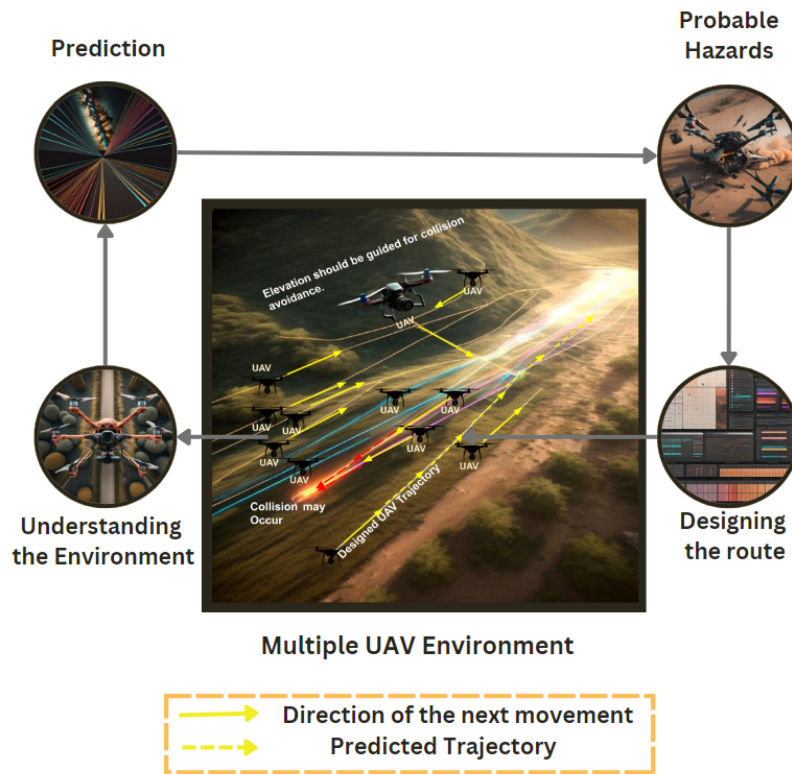


Fig. 4. A Multi-UAV Environment and the Impact of Trajectory Prediction.

connectivity solutions, especially in areas lacking infrastructure or during failures of ground base stations, and the challenges lie in optimizing the 3D locations of UAV-BSs to enhance network performance [89].

6) *Fixed Altitude Assumption*: As shown in [71], [81], [85], [90], [91], earlier works primarily focused on 2D trajectories with fixed UAV altitudes, neglecting the impact of vertical trajectory optimization. The limitations of current models are their heavy dependence on prior knowledge and the lack of multi-step prediction capability [72]. Previous models encountered challenges in handling scale and perspective changes when estimating poses. Different camera angles and distances to gates could lead to errors in pose estimation [92]. Without altitude awareness, UAVs could collide with obstacles such as buildings, trees, or other structures at varying heights within the environment [40].

7) *Limited Battery Capacity*: UAVs have limited battery capacity [93], which affects their deployment and operation, so there is a need for energy efficiency in battery-powered devices [82]. The article [70] discusses that Battery-powered UAVs have limited operational time due to energy constraints, necessitating frequent recharging and impacting sustained communication services. This study aims to optimize resource allocation, considering solar energy, aerodynamic power, and Quality of Service (QoS) constraints. The limitations of limited battery life in UAVs include compromised capability, payload-carrying ability, and control options [94]. Existing methods often needed more consideration for UAV power limitations, obstacle avoidance, and efficient trajectory planning. These limitations included non-holonomic constraints,

exponential time complexity, and neglect of obstacles and power constraints [40].

8) *Uncertain Environments*: Model-based methodologies, including Linear-quadratic (LQ) optimal control and Model Predictive Control (MPC), have shown promise in handling complex systems. However, they rely heavily on accurate system models and can struggle with system nonlinearity and uncertain environments [74]. While intelligent controllers based on machine learning showed promise, ensuring performance and fast convergence with these controllers remained challenging [95]. Previous UAV obstacle detection and avoidance methods involve heavy sensors and complex algorithms, limiting their applicability [96]. Vision-based navigation has emerged as an alternative, but existing approaches suffer from low generalization and reliance on expert demonstrations. Reference [97] introduces a new role, “LifeGuard,” to tackle agents’ apprehensive behavior and improve success rates. Previous trajectory prediction methods, often based on models like Markov Models (MM) and various neural networks, have been applied to land traffic and pedestrian scenarios [98]. However, these models face limitations regarding flight trajectory prediction due to sparse waypoints, changing flight envelopes influenced by external factors, and the inherent complexity of aviation dynamics. This study focuses on improving flight trajectory prediction methods to tackle delays, collisions, and safety concerns while emphasizing the significance of accurate prediction in the context of aviation safety and efficient air traffic management. Reference [99] addresses the challenges of uncertainty in predicting future positions of mobile agents due to both intrinsic randomness (aleatory)

and lack of knowledge (epistemic). Previous methods often focus on modeling motion patterns but neglect the impact of the dynamic environment, leading to limited accuracy. The previous models encountered challenges in multi-agent cooperation due to non-stationary environments caused by policy changes in learning agents [100].

9) *Insufficient Collaboration*: Many models have limitations such as single UAV coverage, insufficient collaboration, overlooking computational tasks, and neglecting latency and interference effects. The study [101] addresses these gaps by proposing a system model with multi-UAV collaborative inference and trajectory planning. The study [102] shows that previous studies have explored various trajectory design objectives but frequently depended on simple Line-of-Sight (LoS) channel models., neglecting the dynamic nature of real-world IoT communication environments. Many existing models needed more consideration of obstacles and practical LoS conditions. Previous models focused on ground vehicles and vessels, benefiting from the structure of road networks and ordered movement patterns. However, UAVs' free movement in three-dimensional space introduces higher uncertainty and complexity [28]. Existing trajectory prediction methods suffer from low accuracy, weak robustness, and limited data scale [103]. The previous machine learning method using Deep Neural Networks for trajectory prediction had limitations in assuming a zero initial velocity for objects in interacting systems, rendering it unsuitable for predicting UAV swarm trajectories in modern warfare scenarios [104].

Summary: The identified challenges show that the previous research works had an emphasis on trajectory planning methodologies. To ensure safe and efficient flight, augmenting trajectory planning with predictive capabilities is imperative [105]. It is noteworthy that trajectory prediction inherently encompasses trajectory planning as an integral component [106]. In this context, the primary objective of this work is to provide a thorough review of methodologies employed in trajectory prediction intricately connected with the trajectory planning. In UAV trajectory prediction, various approaches can be categorized based on their underlying methodologies. Model-Based Approaches encompass two key strategies: Physics-Based Models, which leverage principles of physics and kinematics to predict UAV trajectories, and Analytical Models, employing mathematical equations to describe UAV motion [107]. Data-Driven Approaches leverage historical data, with Statistical Methods using statistical analysis and algorithms, and Machine Learning Techniques employing algorithms like regression, support vector machines, and decision trees. Deep Learning Models, such as recurrent neural networks (RNNs) or LSTM networks, are utilized for intricate pattern recognition [105], [108], [109]. Hybrid Approaches integrate elements from both model-based and data-driven strategies to capitalize on their respective strengths [107]. Trajectory prediction models can be categorized as either Deterministic Models, providing a single predicted trajectory based on known inputs, or Stochastic Models, which account for uncertainty and provide a range of possible trajectories with associated probabilities [32], [33], [74]. The temporal

scope of predictions can be distinguished between Short-Term Prediction, focusing on immediate future trajectories, and Long-Term Prediction, extending forecasts over a more extended time horizon [110], [111]. Additionally, predictions can be made locally, concentrating on limited spatial contexts (Local Prediction), or globally, considering broader spatial scopes (Global Prediction). The temporal nature of predictions is further categorized as either Online Prediction, providing real-time predictions with continuous updates, or Offline Prediction, utilizing historical data without real-time updates [53], [112], [113]. Adaptive Approaches involve adjusting predictions in real-time based on changing environmental conditions [53]. Single-Agent Prediction centers on predicting the trajectory of a single UAV, while Multi-Agent Prediction considers interactions and coordination between multiple UAVs [32], [75]. Finally, trajectory prediction models can be classified as either Probabilistic Approaches, providing probability distributions over possible trajectories, or Non-Probabilistic Approaches, offering a single, deterministic trajectory [114]. This survey addresses gaps in UAV trajectory prediction by categorizing methods, identifying key techniques, analyzing evaluation metrics and datasets, and discussing merits and constraints.

B. Contribution

This survey comprehensively explores trajectory prediction techniques developed for UAVs in recent years. We focus on heuristic and contemporary trajectory prediction methods for comparison and synthesis within a specific time frame.

It is crucial to highlight that historical trajectory data, essential for prediction methods, can be obtained from the perception system [115], [116] and UAV-to-everything (U2X) [117], [118]. Given the undeniable influence of factors like neighboring UAVs on the ego UAV, our primary focus centers on trajectory prediction techniques explicitly tailored for UAVs. In this review, we will assess methods based on Traditional Mathematical Models, Classical ML models, DL methods, and Reinforcement Learning Techniques, respectively, as illustrated in Fig. 5.

This study makes substantial contributions as:

1. A thorough examination of existing trajectory prediction methodologies for UAVs, covering a spectrum from traditional mathematical models and classical machine learning approaches to advanced deep learning techniques and reinforcement learning paradigms.
2. Extensive compilation and analysis of the metrics and datasets used to evaluate the performance of these methods, providing a comprehensive overview of the evaluation landscape.
3. In-depth discussions on the strengths and limitations inherent in each trajectory prediction method, accompanied by insights into promising avenues for future research initiatives in the field.

Unlike previous surveys that often concentrate on a single type of method, our review encompasses a broad spectrum of approaches, from traditional mathematical models and

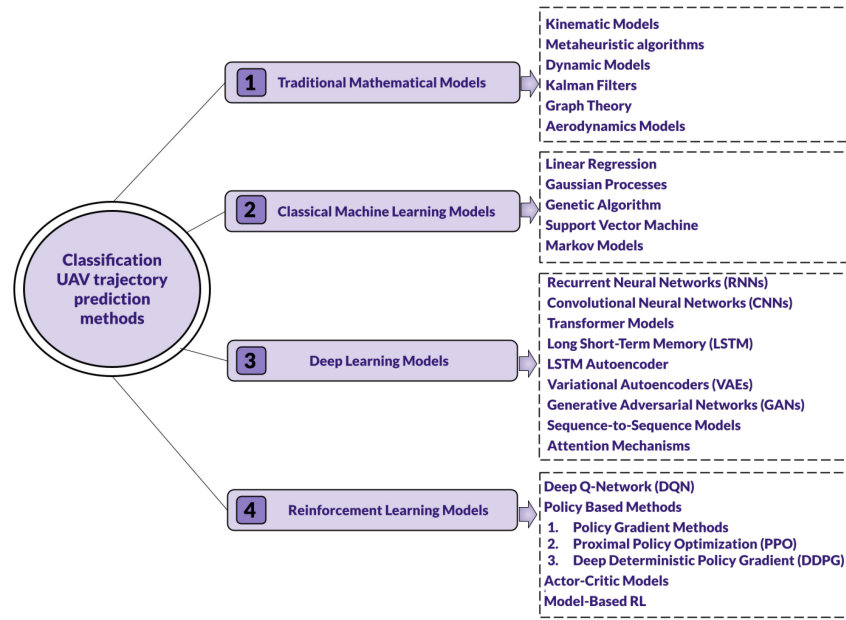


Fig. 5. Classification of Trajectory Prediction Methods for UAVs.

classical machine learning techniques to advanced deep learning and reinforcement learning. Additionally, we offer an extensive analysis of the metrics and datasets employed for evaluating these methods, enriching the evaluation landscape and facilitating better benchmarking.

The following sections are as follows: Section II delineates the trajectory prediction problem and categorizes methods based on various criteria. Sections III–VI conduct in-depth investigations into methods rooted in traditional mathematical models, classical machine learning, deep learning, and reinforcement learning, respectively. Section VII presents an overview of metrics conventionally employed for result comparisons across various studies. Section VIII provides a detail of the datasets utilized in diverse research studies. Section IX offers a comprehensive summary of each method’s strengths and limitations. Considering the limited onboard processing capabilities of UAVs, computational efficiency is a critical factor in evaluating trajectory prediction methods. In this survey, we have also compared the computational efficiency of trajectory prediction methods, in terms of algorithmic complexity, processing time, and resource consumption in Section IX. We summarize findings from benchmarking studies and empirical evaluations to provide a comprehensive overview of how these methods perform under constrained computational environments. The potential avenues for future research are comprehensively outlined in Section X. Finally, our main findings and conclusions are summarized in Section XI.

II. RESEARCH METHODOLOGY AND FORMULATION OF THE PROBLEM

In conducting the research for the UAV trajectory prediction survey, a systematic approach was employed. We began by searching electronic databases, using keywords such as “UAV trajectory prediction” “UAV trajectory design” and

“Unmanned Aerial Vehicle trajectory prediction” This search yielded a total of 994 records from reputable sources, including IEEE, ACM, ScienceDirect(SD), and Springer.

To ensure data quality and prevent duplication, 105 duplicate records were initially removed. Additionally, 75 records were marked as ineligible by automated tools, and 30 records were excluded for various other reasons. Subsequently, the remaining 784 records underwent screening, with 470 excluded based on titles and abstracts. From the remaining 314, 30 couldn’t be accessed. After a closer examination, 152 reports were excluded as they did not meet the predefined inclusion criteria. Finally, the rigorous methodology led to selecting 132 records that were considered pertinent and met the criteria for inclusion in the UAV trajectory prediction survey. This systematic approach ensured the inclusion of high-quality literature in the final review, contributing to the research’s robustness and reliability. An overview of this entire process is visually represented in Fig. 6 for clarity and transparency.

The ‘Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)’ process played a crucial role in the research’s technical contribution by systematically guiding the selection of relevant literature for the UAV trajectory prediction survey. Through a methodical search, screening, and eligibility assessment, it ensured the inclusion of high-quality records, enhancing the robustness and reliability of the overall research.

Further, this section delineates the problem formulation for the UAV location prediction model. The objective is to create a predictive model that incorporates multiple variables, such as the sequence of latitude, longitude, and altitude trajectories, as well as the pressure and temperature data associated with the UAVs. The objective is to predict the next location of the UAV in terms of latitude, longitude, and altitude. The problem of UAV trajectory prediction can be framed as the utilization

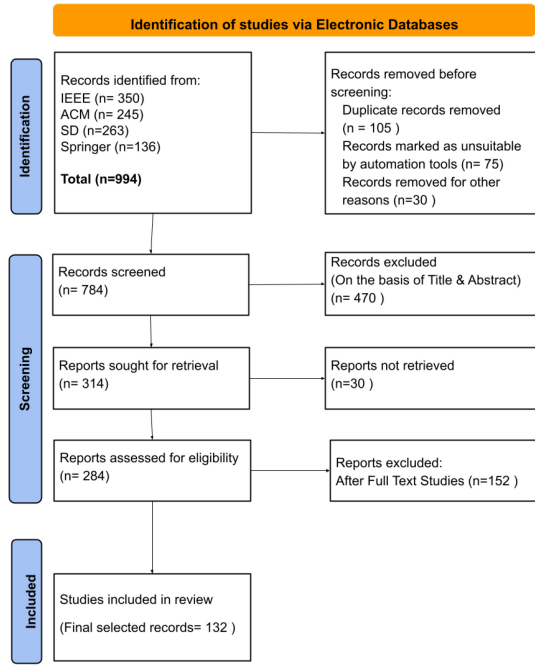


Fig. 6. PRISMA Study Selection Methodology Flowchart.

of previous states of participants within a given scenario to estimate their forthcoming states [98], [119], [120]. The historical states of UAVs, which are participants of air traffic, can be denoted as X :

$$X = \{p_1, p_2, \dots, p_{th}\}, \quad (1)$$

where p_t (for $t \in \{1, 2, \dots, th\}$) represents the states at time step t , th is the length of the historical trajectory, and p_{th} signifies the states of aerial vehicles at the current time or at the time when last location was recorded.

In the majority of trajectory prediction techniques, the variable p_t typically includes the positional information of aerial vehicles and is defined as:

$$p_t = \{(x_{t0}, y_{t0}, z_{t0}), (x_{t1}, y_{t1}, z_{t1}), \dots, (x_{tn}, y_{tn}, z_{tn})\} \quad (2)$$

where n represents all traffic vehicles detected by the UAV, and (x_{tj}, y_{tj}, z_{tj}) denotes the coordinates of vehicle j at time step t .

Input Data: We can represent the input data for the prediction model as X . In this context, X functions as the input to the prediction model, which aims to forecast vehicle trajectories using a time step length of tf . It's worth noting that in certain methods, p_t may encompass supplementary details such as velocity, acceleration, orientation, and so forth.

Output Prediction: The output of the prediction model is denoted as Y , where:

$$Y = \{p_{th+1}, p_{th+2}, \dots, p_{th+tf}\}. \quad (3)$$

The trajectory prediction model is represented as the function F . Some methods can directly generate future trajectories, such as $Y = F(X)$. Others produce intermediate results M , from which Y is derived, i.e., $M = F_1(X)$ and $Y = F_2(M)$. It's crucial to emphasize that M can represent various elements,

including maneuvers generated by maneuver-based techniques or reward functions produced by reinforcement learning-based approaches, among others.

We seek to develop a mathematical model F that maps the input data X to the output prediction Y , i.e., $Y = F(X)$.

The objective of the model is to establish relationships between the input parameters (latitude, longitude, altitude, pressure, temperature) and the UAV's next location. The goal is to create a predictive model that can generalize from the given historical trajectory sequence and environmental data to accurately predict the future UAV location in terms of latitude, longitude, and altitude.

III. TRADITIONAL MATHEMATICAL MODELS

Several traditional mathematical models can be used for UAV trajectory prediction. Some of the commonly used models include:

A. Kinematic Models

These models describe the motion of the UAV based on its position, velocity, and acceleration. They are relatively simple and assume constant velocity or acceleration over short intervals. In [88], the kinematics model is employed to estimate the instantaneous direct and angular accelerations of the flying object based on its estimated state (which includes position, velocity, and possibly other parameters).

The kinematic equations can be adapted to describe the motion of a UAV in three dimensions (x, y, and z coordinates).

Position Equations: The position of the UAV ($\mathbf{r} = [x, y, z]^T$) as a function of time (t) can be expressed using separate equations for each dimension:

$$\begin{aligned} x(t) &= x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \\ y(t) &= y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \\ z(t) &= z_0 + v_{0z}t + \frac{1}{2}a_z t^2 \end{aligned} \quad (4)$$

where, (x_0, y_0, z_0) , (v_{0x}, v_{0y}, v_{0z}) , and (a_x, a_y, a_z) are the initial coordinates, initial velocities and constant accelerations, respectively.

Velocity Equations: The velocity ($\mathbf{v} = [v_x, v_y, v_z]^T$) of the UAV as a function of time can be obtained by taking the derivative of the position equations:

$$\begin{aligned} v_x(t) &= v_{0x} + a_x t \\ v_y(t) &= v_{0y} + a_y t \\ v_z(t) &= v_{0z} + a_z t \end{aligned}$$

Acceleration Equations: If the acceleration is assumed to be constant, then the acceleration ($\mathbf{a} = [a_x, a_y, a_z]^T$) is simply the constant value. Assuming constant accelerations in kinematic models for UAVs is a simplifying but a practical approach employed due to its ease of implementation, computational efficiency, and accuracy within limited time intervals. If accelerations are not constant, the UAV's motion becomes more complex and challenging to predict accurately using simple kinematic models. Therefore, in situations

where accelerations are not constant, more advanced dynamic models adding factors such as aerodynamics, propulsion system dynamics, and environmental conditions are required to accurately capture the UAV's motion.

B. Metaheuristic Algorithms

Optimization algorithms explore solution spaces iteratively to find optimal trajectories [121], examples include Genetic Algorithms and Particle Swarm Optimization etc. The article [82] proposes a non-service-tolerant parameter and introduces a Hybrid Hovering Positions Selection (HHPS) algorithm, employed to identify the optimal hovering positions for UAVs, aiming to minimize the total power consumption of Machine Type Communication Devices (MTCDs) through a trajectory planning approach based on the Cuckoo Search (CS) algorithm. This method focuses on optimizing UAV energy consumption, throughput, and MTCD task latency. The study [21] employs a branch, reduce, and bound (BRB) algorithm along with a low-complexity Successive Convex Approximation (SCA) methodologies to optimize UAV trajectories for timely data collection from IoT devices. The article [122] explores the swarm behavior of small UAVs in pursuit-evasion scenarios. It introduces a physics-based framework for urban environment simulation, enabling quadcopters to interact. A coordinated multiagent system with a target prediction network is proposed for effective pursuit-evasion missions. Unique target prediction is employed, and experiments demonstrate the strategy's performance. The proposed method in [65] considers historical data of the target and the surrounding environment to predict its short-term position while avoiding obstacles. The approach involves a front-end target-informed dynamic searching method and a back-end optimizer to generate collision-free trajectories. In the study [84], Particle Swarm Optimization (PSO) is employed as a method for calculating the optimal flight path of an UAV, collecting data from drifting sea surface sensor nodes.

C. Dynamic Models

These models consider the forces and torques acting on the UAV, including aerodynamic forces, gravity, and thrust. Better suited for scenarios involving changing environmental conditions and aerodynamics. They provide more accurate predictions by considering the UAV's interactions with the environment. In [70], the authors tackle a combinatorial non-convex optimization problem. Their objective is to develop trajectory and resource allocation strategies for solar-powered MC-UAV (Multi-Cell UAV) communication systems. The proposed method [123] is designed to handle the complexities of multi-UAV formation control, particularly in the presence of intricate wind conditions and accounting for uncertainties arising from these wind effects. The study [104] explores employing UAV swarms as interacting systems in combat, emphasizing the need for precise position detection and trajectory prediction. It introduces the Neural Relational Inference (NRI) model with a Mapping Table method linking spring/charged particle models to UAV swarms, culminating in

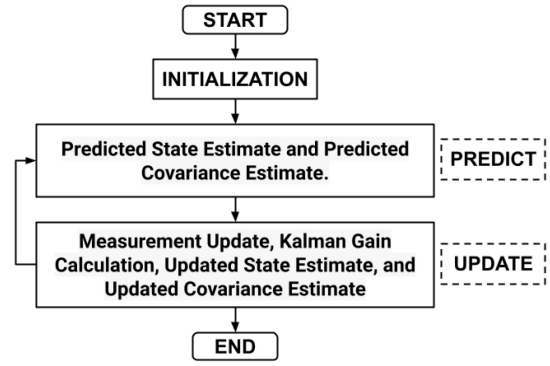


Fig. 7. Working of Kalman Filter Approach.

a novel framework for accurate trajectory prediction substantiated by experiments. The paper [124] presents a Multi-stage Trajectory Design (MSTD) approach for UAVs to optimize joint communication and sensing (C&S) performance. By iteratively refining target location estimates and UAV trajectories, the method balances the tradeoff between total transmitted data for communication users (CUs) and localization accuracy for sensed targets (STs). This dynamic model approach accounts for UAV energy constraints and provides a flexible solution for integrated sensing and communication scenarios.

D. Kalman Filters

Kalman filters combine sensor measurements with predictions from traditional mathematical models to estimate the UAV's current state and improve trajectory prediction accuracy by regularly updating the features [84], [125], working shown in Fig. 7. It is a recursive algorithm that estimates the state of a dynamic system from a series of noisy measurements. It works in the following steps:

1) Prediction Step–State Prediction:

$$\hat{x}_{k|k-1} = F_k \hat{x}_{k-1|k-1} + B_k u_k \quad (5)$$

Error Covariance Prediction:

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q_k \quad (6)$$

2) Update Step–Kalman Gain:

$$K_k = P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} \quad (7)$$

State Update:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (z_k - H_k \hat{x}_{k|k-1}) \quad (8)$$

Error Covariance Update:

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (9)$$

where, $\hat{x}_{k|k-1}$ is the predicted state at time k given measurements up to time $k-1$, $\hat{x}_{k|k}$ is the updated state at time k given measurements up to time k , at time k and F_k is the state transition matrix, B_k is the control input matrix, u_k is the control input, $P_{k|k-1}$ is the predicted error covariance, Q_k is the process noise covariance, H_k is the measurement

matrix, z_k is the measurement, R_k is the measurement noise covariance, K_k is the Kalman gain, and I is the identity matrix.

In UAV trajectory, the state prediction equation given in (5) predicts the UAV's next state using its current state and control inputs; the error covariance prediction shown in (6), estimates the uncertainty in this prediction. The Kalman Gain (7) determines how much to adjust the prediction based on new measurements. The state update using (8) refines the state estimate by incorporating new sensor data, and the error covariance update shown in (9) reduces the uncertainty, improving the accuracy of the UAV's predicted trajectory over time. It is used in the model [88] for denoising and state estimation of the flying object's motion trajectory. In the paper [84] the focus is on UAV path planning for the purpose of collecting data from drifting sea surface sensor nodes. The objective is to find an optimal path that strikes a balance between various factors, including node energy, throughput, flight time, visit frequency, and node position estimation. The approach leverages a Kalman Filter for estimating node motion and employs Particle Swarm Optimization (PSO) for path planning.

E. Graph Theory

They represent UAV trajectories as graphs, where nodes are positions, and edges define possible transitions. Path optimization is done considering spatial relationships and connectivity in complex environments [58], [120]. This study by Zhang et al. [121] optimizes a cellular-enabled UAV system that involves multiple GBSs (Ground Base Stations). The UAV's trajectory aims to maintain connectivity while minimizing mission time adhering to SNR and speed constraints. An efficient algorithm is proposed, leveraging graph theory and convex optimization, yielding close-to-optimal results with reduced complexity. The study [87] describes the problem transformation and introduces two trajectory optimization designs, segment-based trajectory optimization algorithm (STOA) and group-based trajectory optimization algorithm (GTOA), and modeled as the Traveling Salesman Problem with Neighborhoods (TSPN). The Trajectory Construction Subproblem in [63] utilizes the identified strategic points for the UAV to visit for collecting Received Signal Strength Indicator (RSSI) readings from neighboring cells to generate an efficient UAV trajectory. The latter problem is modeled as the Travelling Salesman Problem (TSP), and the 2-opt heuristic solution is used to compute the trajectory efficiently. The study Social-WaGDAT in [120] shows that in the context of trajectory prediction and multi-agent scenarios, graph models are employed to capture the dynamic interactions and dependencies between agents (such as pedestrians, vehicles, or robots) within a scene shown in Fig. 8. Structuring the agents and their interactions as nodes and edges in a graph makes it easier to model and reason about the complex relationships between them.

F. Aerodynamics Models

Aerodynamic models describe the behavior of an object in response to air flow. Different types of models exist

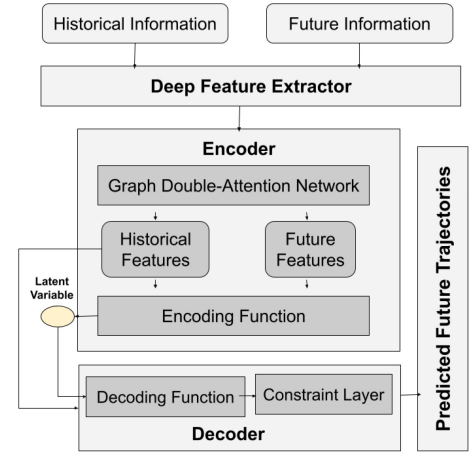


Fig. 8. Structure of Social-WaGDAT for Trajectory Prediction [120].

depending on the complexity of the analysis and the specific characteristics of the object. Aerodynamics-based models are instrumental in UAV trajectory prediction by understanding how UAVs interact with the surrounding air [105], [126], [126]. These models facilitate flight performance predictions, enabling the estimation of critical parameters like lift, drag, and thrust, which, in turn, inform the UAV's speed, endurance, and range [127]. Additionally, aerodynamic models aid in path planning, altitude and speed control, obstacle avoidance, and fuel efficiency optimization by accounting for aerodynamic forces and environmental variables. This knowledge ensures stable and efficient flight operations, making aerodynamics a cornerstone in enhancing the reliability and precision of UAV trajectory predictions [126]. A basic explanation of a simple aerodynamic model can be given in the form of following equations:

1) *Aerodynamic Lift Equation*: The lift force (L) generated by an object, such as an airfoil, can be described by the lift equation:

$$L = \frac{1}{2} \rho v^2 S C_L \quad (10)$$

where, L is Lift force, ρ is Air density, v is Velocity of air relative to the object, S is Reference area (e.g., wing area) and C_L is Coefficient of lift.

2) *Aerodynamic Drag Equation*: The drag force (D) acting on the object is given by the drag equation:

$$D = \frac{1}{2} \rho v^2 S C_D \quad (11)$$

where D is Drag force and C_D is Coefficient of drag.

3) *Aerodynamic Moment (Torque) Equation*: The aerodynamic moment (M) or torque acting on an object can be expressed as:

$$M = \frac{1}{2} \rho v^2 S C_m c \quad (12)$$

where, M is Aerodynamic moment, C_m is Coefficient of moment and c is Reference length (e.g., mean aerodynamic chord).

4) *Coefficient of Lift and Drag*: The coefficients (C_L and C_D) are dimensionless quantities that depend on the geometry of the object and the angle of attack (α):

$$C_L = C_{L0} + \alpha C_{L\alpha} \quad (13)$$

$$C_D = C_{D0} + \left(\frac{\alpha}{\alpha_{\max}} \right)^2 \quad (14)$$

where, C_L , C_D , C_{L0} , C_{D0} , $C_{L\alpha}$, α and $\alpha_{\max}$ are Coefficients of lift, drag, zero angle of attack, at zero angle of attack, Lift-curve slope, Angle of attack and Angle of attack at maximum lift, respectively.

In UAV trajectory prediction models, aerodynamic equations play a critical role. The lift equation (10) helps predict altitude by estimating the lift force, which is influenced by air density, velocity, reference area, and the coefficient of lift. The drag equation (11) predicts the drag force that opposes motion, impacting speed and energy efficiency. The moment equation (12) helps in understanding stability and orientation control. Coefficients of lift and drag (13) and (14) account for changes in the angle of attack, refining the prediction of aerodynamic forces. These equations collectively enable accurate path planning, altitude and speed control, obstacle avoidance, and fuel efficiency optimization in UAV operations.

G. Limitations of Mathematical Models

Predicting the trajectory of Unmanned Aerial Vehicles (UAVs) is a complex task that involves various physical, environmental, and operational factors [128]. While mathematical prediction models can provide valuable insights, they also have several limitations:

1) *Sensitivity to Initial Conditions*: Small errors in initial conditions can lead to significant deviations in the predicted trajectory over time, especially in chaotic or nonlinear systems [129]. UAV trajectories are influenced by wind speed, direction, and turbulence, which can be challenging to measure accurately and account for [4].

For example: Consider a UAV moving vertically along the z-axis with an initial velocity v_{0z} and experiencing a constant acceleration a_z due to gravity. The initial position is z_0 . The kinematic model is described by the equation:

$$z(t) = z_0 + v_{0z}t + \frac{1}{2}a_zt^2 \quad (15)$$

The model has certain restrictions and assumptions:

a. Constant Acceleration: The model assumes a constant acceleration (a_z), overlooking potential variations introduced by factors like air resistance and environment [130].

b. No External Forces Considered: External forces beyond gravity, such as wind and air density, are neglected, impacting the model's accuracy [4], [131].

c. Idealized Initial Conditions: The model assumes precise initial conditions for velocity (v_{0z}) and position (z_0), whereas real-world measurement uncertainties may affect accuracy [21], [70].

d. Linear Motion in One Dimension: The model focuses on linear motion along the z-axis, neglecting potential rotation and motion in multiple dimensions.

This example illustrates the model's limitations, emphasizing the importance of considering more sophisticated models or adjustments when higher accuracy is required [116].

2) *Simplifications and Assumptions*: Mathematical models often rely on simplifications and assumptions to make calculations manageable [132]. However, these simplifications may only capture part of the complexity of UAV flight dynamics, interactions with the environment, and operational constraints.

3) *Limited Environmental Representation*: Mathematical models may struggle to accurately incorporate real-time and dynamic environmental changes, such as sudden wind gusts, temperature variations, and atmospheric disturbances [133], [134]. These factors can significantly impact UAV trajectories but are challenging to predict precisely. Another limitation is the inability to fully capture complex real-world factors such as uncertain wind fields, dynamic environmental conditions, and interactions with other agents [123].

4) *Model Validation and Calibration*: Developing accurate mathematical models requires extensive validation and calibration against real-world data [135]. Unforeseen factors can affect UAV trajectories, making it challenging to validate models comprehensively [136].

5) *Uncertainty and Variability*: Predicting UAV trajectories involves dealing with uncertainties related to sensor measurements, environmental changes, and human inputs [92]. Mathematical models might need to account for these uncertainties, leading to inaccurate predictions adequately. Many models could have struggled to generalize well to various environments and conditions; lighting, weather, and scenery changes could affect the model's performance, leading to poor generalization and reduced robustness [92].

6) *Complex Flight Scenarios*: UAVs often operate in complex and dynamic environments, such as urban areas, where interactions with obstacles, other vehicles, and humans are intricate and challenging to model accurately using mathematics alone [65], [100].

7) *Nonlinear Dynamics*: UAV trajectories can exhibit nonlinear behaviors, especially during maneuvers or when encountering unexpected disturbances. Mathematical models might struggle to capture these nonlinear dynamics accurately [137].

8) *Adaptability*: UAVs may need to adapt their trajectories based on changing mission requirements, unexpected obstacles, or dynamic environmental conditions. Adapting mathematical models on the fly can be challenging and computationally expensive [138].

9) *Data Limitations*: Developing accurate trajectory prediction models requires high-quality training data [135]. In some cases, such data might be scarce or difficult to obtain, limiting the model's accuracy and generalization capability [21], [82], [87], [139], [140].

To address these limitations, a holistic approach that combines mathematical models with real-time sensor data, machine learning techniques, and advanced control algorithms can provide more accurate and reliable trajectory predictions for UAVs. Table II discusses the main approaches used in trajectory prediction using traditional mathematical models.

TABLE II
THE PREVALENT STRATEGIES EMPLOYED USING MATHEMATICAL MODELS

Author	Year	Method	Dataset	Main Findings	Evaluation Metrics	Limitations	Future Scope
Zhang et al. [121]	2018	Graph Convex optimization, and Dijkstra's algorithm	Simulation-based	Optimizes UAV trajectories for cellular communication, achieving connectivity while minimizing mission time and complexity	Completion Time, Signal-to-Noise (SNR)	Focuses on single UAV scenarios; extension to multiple UAVs might introduce complexities.	Explore multi-UAV coordination strategies to enhance system efficiency.
Zhu et al. [82]	2018	Cuckoo Search	Simulation: Geographical area: 500m x 500m using a cluster-based PPP (Poisson Point Process) distribution	Utilizing a trajectory planning method based on the Cuckoo Search algorithm has resulted in enhanced energy efficiency.	Battery Consumption of UAV, Total Latency of MTCD Tasks, UAV Throughput	The current spatial resolution (13km) limits higher horizontal accuracy, requiring improved weather data.	Investigate higher spatial resolution weather data for improved trajectory prediction accuracy.
Li et al. [87]	2019	STOA, GTOA, Attitude Optimization, Joint Optimization	N Ground users (GUs) distributed uniformly within a 3000 m x 3000 m square area.	(STOA and GTOA) and Optimizations aim to achieve more efficient data collection using UAVs.	Computation Time Comparison	Assumes ideal air-to-ground channel conditions, lacks consideration for real-world interferences and dynamic environments.	Investigate hybrid optimization techniques and real-time adaptability for varying UAV missions and environments.
Sun et al. [70]	2019	Non-convex optimization problem	Simulations: One circular cell, K users, 800m radius.	Optimizes trajectory, power, and subcarrier allocation to maximize system throughput over a finite horizon for solar-powered MC-UAV systems.	Convergence and Accuracy Analysis	Neglects inter-UAV interference effects in dense scenarios.	Investigate adaptive energy-efficient trajectory strategies considering real-time cloud coverage.
Samir et al. [21]	2020	Branch, Reduce and Bound (BRB)	Simulations: geographical area of 0.8 km by 0.8 km, assumes multiple IoT devices	A Rician channel model to ensure optimally designed trajectories, collect data from the maximum number of devices maintaining a minimum data upload.	Percentage of served IoT devices.	Limited to OMA, lacks consideration of NOMA-based UAV scenarios, and energy consumption analysis.	Joint optimization of trajectories and radio resources for multiple UAVs.
Zhang et al. [123]	2020	A cooperative guidance control method is proposed based on the backstepping approach.	Simulation: MATLAB R2014a and Simulink for virtual simulation experiments	Backstepping control method are shown to enable each UAV to rapidly converge to the desired trajectory, achieving good tracking performance and stability in UAV formation control.	Convergence Analysis, Robustness Analysis	Limited to takeoff-to-assembly process, lacks task-specific considerations and neglects closed formation vortex and wind field effects.	Extend to task-oriented formation control, consider UAV mission, shape, and quality, and account for closed formation vortex and wind field effects.
Wang et al. [104]	2020	Neural Relational Inference Model	Three UAV swarm datasets are generated based on the Mapping Table.	The paper introduces a novel framework for predicting the trajectory of UAV swarms.	Accuracy	Limited to two-dimensional projection, potential accuracy improvement in three-dimensional trajectory prediction.	Investigate three-dimensional trajectory prediction for UAV swarm using the proposed framework.
Cui et al. [65]	2022	Heuristic function for dynamics search	Simulation utilized a detailed city model with various non-convex obstacles	achieves active and safe target tracking through a secure trajectory planning strategy and integration into a UAV system.	Accuracy.	Difficulty in predicting extreme target motions in open environments; addressed by the path searcher.	Explore multi-UAV tracking of a single target, formation flight of drones, and improved object recognition.
Jing et al. [124]	2024	MSTD and successive convex approximation	Simulated data based on UAV communication and sensing scenarios	MSTD approach balances the tradeoff between communication and sensing performance, improving total transmitted data and localization accuracy.	Total data, Bound, Cramér-Rao	optimization problem is solved for local optima, which may not guarantee a global optimality.	Incorporating more realistic environmental models and real-world datasets for validation.

In the presented research summary (Table II), various studies are summarized based on key features. The 'Author' column identifies the primary contributors of each study. The 'Year' column indicates the publication year. The 'Method' column outlines the distinct approaches employed in each study, showcasing the diversity of methodologies in trajectory prediction research. The 'Dataset' column specifies the datasets utilized. The 'Main Findings' column succinctly encapsulates the

primary discoveries and contributions made by each study. Evaluation metrics are detailed in the 'Evaluation Metrics' column, shedding light on the quantitative measures used to assess the methods' performance. 'Limitations' in the subsequent column enumerate the constraints and challenges faced by each method, enhancing transparency. Finally, the 'Future Scope' column outlines potential directions for future research.

IV. CLASSICAL MACHINE LEARNING MODELS

Several ML models can be used for UAV trajectory prediction, depending on the complexity of the problem and the available data [11], [141], [142]. A detailed description of some popular methods has been given in Table III. Here are some commonly used machine-learning models for UAV trajectory prediction:

A. Linear Regression

Linear Regression is a fundamental technique in UAV trajectory prediction, providing a simple yet effective model for capturing relationships between input features and output trajectories [143], [144]. In the context of trajectory prediction, Linear Regression aims to find the best-fitting linear relationship. The key components and equations for Linear Regression can be expressed as follows:

Linear Regression Equation:

$$f(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$$

Here, $f(\mathbf{x})$ represents the predicted trajectory, $\mathbf{w}$ is the weight vector, $\mathbf{x}$ denotes the input features, and b is the bias term.

Objective Function (Least Squares):

$$\min_{\mathbf{w}, b} \sum_{i=1}^n (y_i - (\mathbf{w} \cdot \mathbf{x}_i + b))^2$$

Minimizing the sum of squared differences between predicted ($\mathbf{w} \cdot \mathbf{x}_i + b$) and actual trajectories (y_i).

Closed-Form Solution (Normal Equation):

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$b = \bar{y} - \mathbf{w} \cdot \bar{\mathbf{x}}$$

Here, $\mathbf{X}$ is the matrix of input features, $\mathbf{y}$ is the vector of output trajectories, $\bar{\mathbf{x}}$ is the mean of input features, and $\bar{y}$ is the mean of output trajectories.

Linear Regression provides a straightforward approach for UAV trajectory prediction, with model parameters determined analytically. While simplistic, it serves as a baseline for comparison with more complex models in trajectory prediction applications.

The study [139] involves optimizing the UAV's 3D trajectory while considering angle-dependent Rician fading channels for communication with Sensor Nodes (SNs) on the ground. This approach includes constraints on UAV flying speeds, initial/ final locations, minimum altitude, and specific fading channel characteristics. The optimization is carried out by employing a convex solver that relies on the interior-point method. The effective fading power parameter is approximated using a logistic function, which is a common approach in regression analysis.

Simple linear regression can be used when there is a linear relationship between input features (such as time, velocity, and sensor data) and the predicted trajectory [144]. The method in [143] used for UAV path prediction in this study, a regression algorithm based on Recursive Area Clustering, is employed. This algorithm involves the identification of high-density areas extracted from the historical trajectories of

TABLE III
THE PREVALENT STRATEGIES EMPLOYED USING MACHINE LEARNING-BASED METHODOLOGIES

Author	Year	Method	Dataset	Main Findings	Evaluation Metrics	Limitations	Future Scope
Ayhan et al. [151]	2016	Stochastic HMM	Real host track and aircraft specification data	HMM-based system: benefiting tactical aircraft trajectories for ground-based operations.	Trajectory prediction accuracy metrics	The current spatial resolution (13km) limits higher horizontal accuracy, requiring weather data.	Investigate higher spatial resolution weather data for improved trajectory prediction accuracy.
Wang et al. [12]	2017	Hybrid Clustering and Multi-Cells Neural Networks (MCNN)	ADS-B records: July 2017, Beijing Capital International Airport (BCIA) TMA, 3 parallel runways: 18R/36L, 18L/36R, and 01/19.	Enhances the precision of 4D trajectory prediction.	MAE, RMSE	Restricted to Expected Time of Arrival (ETA) prediction; needs investigation on diverse trajectory variables and model comparisons.	Investigate varied lookahead times, contrast model-based techniques, examine diverse trajectory factors.
You et al. [139]	2019	Optimizes communication scheduling and trajectory.	UAV-enabled WSN scenario with SNs randomly and uniformly distributed in a square area	Constrained 3D Trajectory Optimization with Angle-Dependent Rician Fading Modeling for UAV-enabled data harvesting	Max-min rate and estimated max-min rate	Lack of closed-form for fading power, simplification of LoS, 3D trajectory complexity.	Extend to multi-UAV networks, optimizing 3D trajectories, communication, and resource allocation
Aslan et al. [68]	2022	VTONet, is proposed for UAV pose estimation	EuRoC dataset	Introduces a unique strategy for UAV pose estimation that fuses OF-Inertial-Visual features, outperforming previous methods, utilizing a combination of optical flow, inertial data, and visual information.	Pose estimation accuracy	Limited to monocular camera and IMU; not tested in outdoor scenarios, potential overfitting with GPR.	Explore semantic SLAM integration, handle dynamic environments, enhance pose estimation robustness, optimize CNN architecture.
Zhang et al. [29]	2022	PCA-SVM	GPS data, F450: UAV.	Introducing flight state recognition to deep learning-based UAV trajectory prediction enhances accuracy by accounting for flight state effects on trajectories.	MSE, Regression Coefficient	Requires human expertise for flight state determination; limited to specific experimental settings; may benefit from more diverse datasets.	Explore UAV trajectory prediction in cluster collaboration and incorporate deep reinforcement learning for improved flight state recognition.

drifting targets. Subsequently, strategic zones (weight centers) are determined, and the object's trajectory is predicted using refined regression based on these weight centers. The predicted trajectory is then utilized to plan the path for the UAV search mission.

B. Gaussian Processes

Gaussian process models can capture uncertainties in trajectory predictions and provide probabilistic forecasts as mentioned in [145], which focuses on Gaussian process-based channel prediction specifically tailored for communication-relay UAVs operating in urban environments.

Gaussian Processes (GP) play a pivotal role in UAV trajectory prediction, employing probabilistic modeling techniques. GPs provide a flexible and Bayesian approach for capturing uncertainties and making predictions. In the context of trajectory prediction, GPs are utilized to model the distribution of possible trajectories [68], [139]. The key components and equations for Gaussian Processes can be expressed as follows:

1. Prior Distribution:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (16)$$

Here, $m(\mathbf{x})$ is the mean function, and $k(\mathbf{x}, \mathbf{x}')$ is the kernel function.

2. Kernel Function:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right) \quad (17)$$

A common choice is the radial basis function (RBF) kernel.

3. Training Data:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n] \quad \text{and} \quad \mathbf{y} = [y_1, y_2, \dots, y_n]$$

Representing the input-output pairs used to train the Gaussian Process model.

4. Posterior Distribution:

$$f_*|\mathbf{X}, \mathbf{y}, \mathbf{x}_* \sim \mathcal{N}(\mu_*, \Sigma_*) \quad (18)$$

where μ_* is the predictive mean, and Σ_* is the predictive covariance.

5. Predictive Mean and Variance:

$$\mu_* = k(\mathbf{x}_*, \mathbf{X}) \left(K + \sigma_n^2 I \right)^{-1} \mathbf{y} \quad (19)$$

$$\Sigma_* = k(\mathbf{x}_*, \mathbf{x}_*) - k(\mathbf{x}_*, \mathbf{X}) \left(K + \sigma_n^2 I \right)^{-1} k(\mathbf{X}, \mathbf{x}_*) \quad (20)$$

Here, K is the covariance matrix for the training data, and σ_n^2 is the noise variance.

By using the probabilistic nature of GPs, UAV trajectory predictions can account for uncertainties and provide reliable estimates. The specific implementation may involve choosing appropriate kernel functions and training the model with trajectory data specific to the UAV application. The research [68] focuses on estimating the position of UAVs by combining visual-inertial data by converting inertial data into images and merging it with optical flow frames and employing Inception-v3 for feature extraction and Gaussian Process Regression for pose estimation, providing stable results on actual and simulated datasets. In the study [99], Gaussian distributions are used to represent uncertainty and variability in the predictions of the future position of a target object. Gaussian distributions are commonly employed to model uncertainty in various domains due to their mathematical properties and the Central Limit Theorem. The proposed algorithm [92], integrating

Dynamic Domain Rapidly-Exploring Random Tree (DDRRT) and Linear Quadratic Gaussian Motion Planning (LQG-MP), demonstrates effectiveness in safely navigating through uncertain and hostile environments considering threats, vehicle motions, and observations based on Monte Carlo simulations.

C. Genetic Algorithm

Genetic Algorithm-based Machine Learning models, enhance UAV trajectory prediction by optimizing flight paths and control parameters through iterative selection and adaptation, improving trajectory efficiency [111], [146], obstacle avoidance, and adaptability to dynamic conditions [111], ultimately enhancing mission success and precision. The study [111] shows a short-term 4D trajectory prediction model for UAVs by leveraging spatio-temporal trajectory clustering techniques. The goal is to enhance accuracy and reliability in UAV trajectory prediction.

Genetic Algorithms (GA) play a pivotal role in UAV trajectory prediction, employing specific optimization techniques. GAs are heuristic search algorithms inspired by the process of natural selection, aiming to evolve solutions for complex problems. In the context of trajectory prediction, GAs are utilized to optimize UAV paths based on predefined objectives [111], [146], [147]. The key components and equations for GAs can be expressed as follows:

Genetic Algorithm Components:

1. **Chromosomes:** A representation of potential solutions, often encoded as binary strings.

2. **Population:** A collection of individual solutions or chromosomes.

3. **Fitness Function:** A metric that evaluates the performance of each solution based on predefined criteria.

4. **Crossover (Recombination):** A genetic operation that combines two parent solutions to create new offspring solutions.

5. **Mutation:** A genetic operation that introduces random changes in a solution to explore new possibilities.

6. **Selection:** The process of choosing individuals from the population for reproduction based on their fitness.

Genetic Algorithm Workflow: Repeat until convergence: Select, Crossover, Mutate, Evaluate Fitness

These components define the core elements of a genetic algorithm. By iteratively evolving populations of solutions, GAs can optimize trajectories for UAVs. The specific implementation may vary based on the trajectory prediction problem and optimization objectives. It's crucial to customize GAs to suit the characteristics of trajectory data and the specific requirements of the UAV application. The study [146] utilizes a Multi-Population Genetic Algorithm (MPGA). Firstly, MPGA employs multiple populations with different control parameters to conduct parallel optimization searches simultaneously. Second, each population operates relatively independently but is interconnected through migration operators, ensuring coevolution and preventing the loss of optimal solutions. Third, MPGA introduces an elite population, separate from the general population, to record and retain the best solutions from all populations during iterations, aiding

algorithm termination criteria and preserving optimal solutions. The study [147] employs genetic algorithms to create a 4-D flight-trajectory prediction model for UAVs, considering longitude, latitude, altitude, and entry time into protection zones. The model also extracts parameters like entry time and crossing-point number to assess dangerous behavior intentions accurately, ensuring operational safety. The article [148] introduced a model utilizing genetic algorithms for predicting transport aircraft's climb and descent trajectories.

D. Support Vector Machine

Support Vector Machines (SVM) in the context of UAV trajectory prediction involve the use of specific mathematical equations. The SVM algorithm aims to find a hyperplane that best separates data points into different classes. In a binary classification scenario, the equation for a linear SVM can be expressed as follows:

Linear SVM Equation:

$$f(x) = \mathbf{w} \cdot \mathbf{x} + b \quad (21)$$

Here:

- $f(x)$ is the decision function.
- $\mathbf{w}$ represents the weight vector.
- $\mathbf{x}$ denotes the input features.
- b is the bias term.

For non-linear SVMs, the kernel trick is commonly employed to map the input features into a higher-dimensional space. The equation for a non-linear SVM with a kernel function (K) can be expressed as:

Non-Linear SVM Equation:

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \quad (22)$$

Here:

- α_i represents the Lagrange multipliers.
- y_i is the class label of the training sample $\mathbf{x}_i$.
- $K(\mathbf{x}_i, \mathbf{x})$ is the kernel function that computes the dot product in the higher-dimensional space.

Popular kernel functions include linear, polynomial, and radial basis function (RBF) kernels.

The model [29] utilizes a Principal Component Analysis-Support Vector Machine (PCA-SVM) for recognizing flight states and employs a neural network for trajectory prediction based on the identified states.

The work utilizes Support Vector Machine (SVM) as part of its methodology for flight state recognition. SVM is employed after Principal Component Analysis (PCA) to classify different flight states of the UAV based on the collected flight data. The PCA-SVM model aids in accurately identifying the current state of the UAV, which is crucial for subsequent trajectory prediction. By incorporating SVM into the workflow, the model enhances its ability to classify the UAV's flight states with high accuracy, thereby improving the overall performance of trajectory prediction.

Experimental outcomes demonstrate precise state recognition, with an average prediction error of 0.214 meters for the UAV flight path, surpassing conventional techniques such

as the Unscented Kalman Filter. The study [149] is using UAV PCA-DAGSVM model for flight state classification. DAGSVM (Directed Acyclic Graph Support Vector Machine) arranges multiple binary SVM classifiers in a directed acyclic graph structure, allowing for effective multi-class classification. Although, the trajectory prediction has been performed using neural networks.

E. Markov Models

Markov models utilize probability distributions to predict UAV positions based on their current states. Hidden Markov Models (HMMs) are particularly crucial in this context, as they employ specific mathematical equations to model the underlying dynamics of systems with hidden states. For UAV trajectory prediction, HMMs capture the latent variables that influence UAV movement, providing a robust framework for accurately forecasting future positions.

The key equations for HMMs can be expressed as follows:

1. State Transition Probability:

$$P(q_t | q_{t-1})$$

Probability of transitioning to state q_t from q_{t-1}

2. Observation Probability:

$$P(x_t | q_t) \text{ Probability of observing data } x_t \text{ given the state } q_t$$

3. Initial State Probability:

$$P(q_1) \text{ Probability distribution of the initial state } q_1$$

These equations define the core components of an HMM. Here, q_t represents the hidden state at time t , and x_t denotes the observed data at time t . By utilizing these probabilities, HMMs can infer hidden states and make predictions about future trajectories based on observed data. Markov models utilize probability distributions to predict UAV positions based on their current states. Hidden Markov Models (HMMs) extend this by using specific equations to model the dynamics of systems with hidden states, crucial for accurate UAV trajectory prediction. The state transition probability $P(q_t | q_{t-1})$ models the likelihood of moving from the previous state q_{t-1} to the current state q_t . The observation probability $P(x_t | q_t)$ relates observed data x_t to the current state q_t , allowing inference of the UAV's state from measurements. The initial state probability $P(q_1)$ sets the initial condition for the model. Using these probabilities, HMMs capture latent variables influencing UAV movement, enabling precise prediction of future positions and enhancing UAVs' operational efficiency and reliability.

HMMs can account for uncertainty in UAV motion. The research by Ayhan and Samet [150] presents three key contributions: a novel representation of aircraft trajectories using 4D joint cubes formed through alignment and fusion, introduces Stochastic HMM with historical data and weather for better trajectory prediction. Use time series clustering for realistic flight plans. A Markov decision process model in [75] combines long-distance autonomous guidance and short-distance tracking for UAV obstacle avoidance with an enhanced Proximal Policy Optimization (PPO) algorithm that

uses an LSTM network to optimize UAV maneuvering based on real-time position and contextual data, achieving efficient and robust autonomous navigation.

F. Limitations of Machine Learning Models

1) *Data Availability and Quality*: Accurate trajectory prediction requires high-quality training data. Such data can be challenging and expensive, especially for rare or critical events [151]. Obtaining comprehensive, diverse, and representative real-world data can be challenging. Inadequate or biased data can lead to poor model generalization and accuracy. Moreover, more access to high-quality data for various flight scenarios and conditions is needed to develop robust and reliable UAV trajectory prediction models [152].

For example: Limited historical data for UAV flights in remote regions with unique weather conditions may hinder accurate trajectory predictions for rare or specific flight patterns.

2) *Generalization*: ML models trained on specific datasets may struggle to generalize to new environments, weather conditions, or scenarios not covered in the training data. The models' performance relies on the training data's quality and diversity. Using different drone platforms or operating conditions might reduce prediction accuracy if the training data is not representative [113]. For example: An UAV trajectory model trained in urban landscapes might struggle to generalize effectively to rural or mountainous terrains due to differences in structures and topography.

3) *Dynamic Environments*: ML models might struggle to adapt to rapidly changing environments, such as crowded urban areas or airspace with multiple moving obstacles. The machine learning models are designed to predict flight time along predefined paths. They may not be suitable for other types of predictions or broader applications [113]. ML-based solutions, including Q-learning and prediction-based methods, lack adaptability to changing environments and focus on prediction with less training data rather than trajectory design [86]. Traditional generative models like Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) face challenges in accurately predicting trajectories for motion prediction tasks. GANs often suffer from mode collapse, failing to capture the complete data distribution. VAEs approximate the true distribution with a lower bound and require computationally expensive Monte Carlo methods for estimating likelihoods [153].

4) *Limited Understanding of Physics*: Many ML models lack an understanding of physical laws and constraints, which can lead to physically unrealistic or unsafe trajectories [36]. These models rely heavily on data patterns and may overlook intricate aerodynamic principles and environmental effects. Consequently, they may need to account for factors like wind, turbulence, or unusual flight conditions, leading to less reliable predictions, especially in complex scenarios requiring precise and physics-based trajectory forecasting [77], [78].

5) *Uncertainty Handling*: Most ML models provide deterministic predictions, but in real-world scenarios, uncertainty is prevalent due to sensor noise, occlusions, and unpredictable

agent behaviors [34]. They may need the ability to effectively quantify and manage uncertainty associated with factors like wind, turbulence, or unpredictable environmental conditions. This limitation can result in inaccurate predictions, potentially compromising the safety and reliability of UAV operations, especially in scenarios where precise trajectory forecasting is essential [18], [92], [154]. For example: UAV trajectory models may struggle to account for uncertainties in atmospheric conditions, leading to less reliable predictions during turbulent weather encountered in coastal regions.

6) *Real-Time Constraints*: UAV trajectory prediction often requires real-time decision-making. Complex ML models might need to meet the computational efficiency required for real-time operation [155]. ML models can be computationally intensive, causing delays in decision-making during critical situations. This latency can impact the UAV's ability to adapt to rapidly changing environments or make timely course adjustments, potentially compromising safety and mission success, especially in scenarios requiring quick and responsive trajectory predictions [112]. For example: In time-sensitive applications like search and rescue missions, UAV trajectory models with long computation times may impede swift decision-making during critical situations.

7) *Data Imbalance*: In scenarios with rare events (e.g., emergencies), training data might be imbalanced, leading to biased prediction [21], [78]. When datasets need balance, models can become biased towards the majority class, resulting in poor performance for rare or underrepresented flight scenarios [102]. This may lead to inaccurate trajectory predictions, especially when the model needs more exposure to less common environmental conditions or flight patterns, impacting overall prediction reliability [156].

8) *Interactions and Cooperation*: ML models might struggle to capture complex interactions and cooperative behaviors among multiple agents in a shared airspace [157]. They often oversimplify these dynamics, hindering the accurate prediction of joint behaviors or coordinated actions. This can lead to suboptimal trajectories and challenges in scenarios where multiple UAVs must work together, like search and rescue missions or surveillance operations, potentially compromising mission effectiveness and safety [158].

9) *Extrapolation*: ML models can struggle with extrapolation beyond the training data's temporal and spatial boundaries, leading to unreliable predictions in unfamiliar situations [111]. These models may need help to make accurate predictions in unfamiliar or unanticipated environments that lie beyond their training data.

Consequently, their reliability diminishes in scenarios with unique or uncommon conditions, potentially leading to inaccurate trajectory forecasts and performance degradation in real-world UAV applications [159].

10) *Adversarial Attacks*: ML models can be sensitive to adversarial attacks, which involve making tiny, often imperceptible changes to input data. These subtle alterations can cause ML models to make big mistakes in their predictions, potentially leading to unsafe trajectories [160]. These attacks can manipulate input data to deceive the model into making incorrect predictions, posing security risks [161]. Adversarial

attacks can compromise UAV safety and privacy, potentially leading to dangerous flight paths or unauthorized access to sensitive information, making it essential to bolster model defenses against such threats.

V. DEEP LEARNING MODELS

In trajectory prediction using deep learning, models like Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) utilize previous trajectory data as input, capturing temporal dependencies. Fig. 9 shows the working of CNN and Fig. 10 provides a visual representation of how these models process historical sequences to predict future trajectories, highlighting their ability to learn and generalize temporal patterns. Several deep learning models can be used for UAV trajectory prediction owing to their capacity to capture intricate patterns in sequential data such as [27], [28], [30], [38], [42], [72], [98], [108], [109], [110], [119], [162], [163], [164], also discussed in Table VI. Here are some deep learning models commonly used for UAV trajectory prediction:

A. Recurrent Neural Networks (RNNs)

The trajectory prediction in [27] employs an RNN-based model that predicts a distribution of the successive N image positions based on previous observations as shown in Fig. 10. The model utilizes a Mixture Density Network (MDN) and captures motion variations for different object types. RNNs are designed to work with sequential data and are suitable for trajectory prediction tasks [98]. The article [31] proposes a grouping-based conflict detection algorithm and combines it with RNN to predict short-term and long-term trajectories. The study [119] introduces a novel approach that combines CNN and RNN to extract both spatial and temporal dimensions of aircraft trajectories, addressing the limitations of previous models that primarily relied on RNN alone. In [162], DL (Deep Learning) approaches, including MLP (Multi-Layer Perceptron) and LSTM, are proposed for predicting UAV positions and maximizing user throughput. Using trained neural networks, [113] estimates times for standard sub-paths, then sums them for total flight time. Accurate drone tests validate the method's accuracy, which aids collision prediction in unmanned traffic management.

B. Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) process visual data by using convolutional layers that apply filters to input images, capturing hierarchical features. Pooling layers reduce spatial dimensions, and fully connected layers interpret the learned features for classification or regression tasks as shown in Fig. 9. Although CNNs are frequently employed for image data, they can also be applied to trajectory prediction tasks involving spatial and temporal features. They might be helpful when trajectory data is represented in a grid-like format. The study [39] emphasizes the generation of realistic wind data for testing small UAVs in urban environments. Large Eddy Simulations (LES) data and Convolutional Auto-Encoders are combined to predict wind field data.

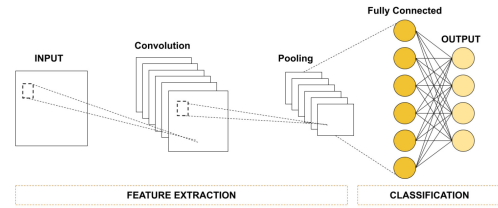


Fig. 9. Architecture of CNN.

High-fidelity LES data trains machine learning techniques for generating wind data in complex urban settings. The proposed method in [69], called ‘GreySeqNet’ employs a compact convolutional neural network using grayscale images, processing them in sequences to achieve high-frequency operation. This approach aims to address the trade-off between localization accuracy and processing time.

C. Transformer Models

Originally developed for natural language processing, Transformer models can also be adapted for trajectory prediction. They excel at capturing long-range dependencies in sequential data and have demonstrated success in various sequence-to-sequence tasks [164]. The proposed approach in [38] combines the closed-loop prediction, DL-based Trajectory-Mapping Network (TMN) and the Time-series Convolutional Neural Network (TSCNN), and optimization under model predictive control (MPC) for highly feasible UAV trajectory planning. The Latent Variable Sequential Set Transformers [165], known as “Autobots,” propose encoder-decoder architectures for robust multi-agent trajectory prediction by effectively capturing contextual, social, and temporal information. These architectures utilize interleaved temporal and social multi-head self-attention modules in the encoder, facilitating scene-consistent trajectory generation. In the decoder, learnable seed parameters are combined with temporal and social modules to enable efficient inference over the future scene. The model [166] addresses privacy concerns with increasing UAV use; the abstract introduces ConvLGBM, a novel technique merging Convolution Neural Network (CNN) feature extraction and Light Gradient Boosting Machine (LightGBM) classification for effective drone detection across a range of Radio Frequencies (RF). The study [96] presents a self-supervised CNN-based method for autonomous indoor robot navigation, with a focus on obstacle avoidance. It employs a regression CNN to predict collision proximity using raw visual data from the robot’s camera, trained on an annotated indoor-flight dataset with real-distance labels. This allows the robot to autonomously navigate and avoid obstacles based on predicted collision distances. Transformer models, known for their ability to capture complex sequential dependencies, have shown promise in this context. This characteristic makes them well-suited for the prediction of complex sequences like the remaining useful life (RUL) of highly maneuverable UAVs considering future mission maneuvers [94]. The study [99] employs convolutional neural networks and Gaussian distributions to enhance trajectory planning for mobile robots, tested using the Stanford

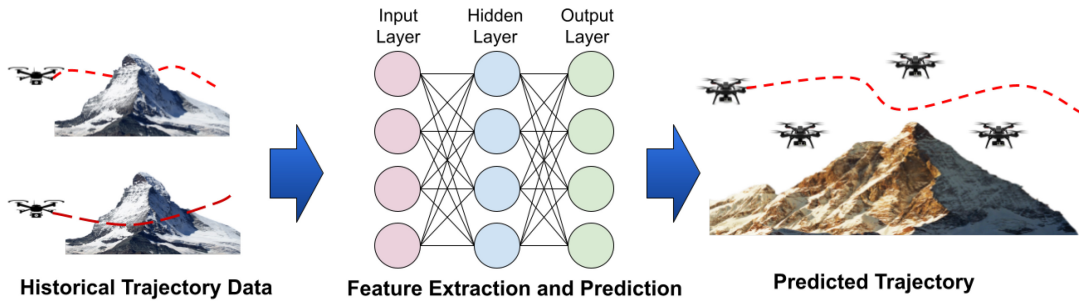


Fig. 10. Working of Deep Learning based Models.

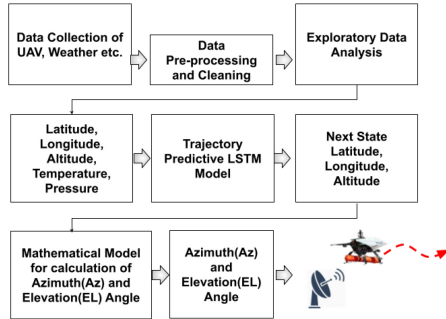


Fig. 11. Working of LSTM Autoencoder Method for Trajectory Prediction including Azimuth and Elevation Angles [105].

Drone Dataset. It introduces a deep learning method that utilizes CNNs and unsupervised clustering to predict multiple hypotheses for the future locations of a target object. The study [167] introduces DeepPilot, a CNN model, to predict flight commands using images; it enables drones to navigate through race gates independently, utilizing deep learning for better performance. The paper [110] presents MALS-Net, a novel approach that combines LSTM and multi-head attention mechanisms, capitalizing on transformer-inspired techniques. Unlike previous methods reliant on datasets prone to noise and overfitting, MALS-Net is evaluated on a more practical dataset, BLVD: Building A Large-scale 5D Semantics Benchmark for Autonomous Driving. The research [168] suggests a vision-based control system for UAV to Unmanned Ground Vehicle (UGV) interaction. It predicts UGV motion and enables agile UAV maneuvers. The study [169] proposes a time-varying nonholonomic UAV formation control method using nonlinear model predictive control (NMPC) combined with discrete control barrier functions (DCBF) and a transformer network for obstacle trajectory prediction, ensuring optimal obstacle avoidance and smooth control inputs. The methodology is verified through numerical simulations, demonstrating its effectiveness in reducing tracking errors and ensuring safe UAV formations.

D. Long Short-Term Memory (LSTM)

These models combine the sequence modeling capabilities of LSTMs with autoencoders for feature extraction, particularly useful for sequential trajectory data. The paper [98] presents an emphasis on precise flight trajectory prediction

TABLE IV
COMPARISON OF EXISTING MODELS WITH LSTM-AUTOENCODER
AS GIVEN IN [105]

Model	MSE	RMSE
Multi-output Regression	226.65	15.05489
Vanilla LSTM	127.76	11.30309
LSTM Auto-encoder	0.0121	0.11040

within air traffic management, utilizing a LSTM network-based model to forecast flight paths, encompassing both 3D and 4D data dimensions. The method employs sliding windows within the LSTM framework to ensure trajectory continuity, enhance prediction accuracy, and evaluate the model in terms of Mean Absolute error (MAE), Mean Relative Error (MRE), Root Mean Square Error (RMSE), and Dynamic Warping Time (DWT). Given the challenges of UAV trajectory prediction in dynamic flight scenarios, the RNN with LSTM is employed in [28]. The proposed algorithm, named Recurrent-LSTM (RLSTM), uses Automatic Dependent Surveillance–Broadcast (ADS-B) data for accurate prediction. Simulations demonstrate that the RLSTM-based approach effectively predicts UAV trajectories with satisfying accuracy, addressing the challenges of air traffic control for collision avoidance among UAVs. The paper [103] introduces a method involving time difference information from electromagnetic signals and the Chan method for trajectory positioning alongside a long-term memory neural network for UAV trajectory prediction, aiming to enhance accuracy and provide a novel solution for trajectory analysis. The study [108] discusses safety in autonomous UAV missions; it introduces a deep neural network model, Stacked Bidirectional and Unidirectional LSTM (SBULSTM), to predict UAV positions based on past data. Predictions guide safe UAV control during cruise missions, ensuring completion within a set threshold.

E. LSTM Autoencoders

These models combine the sequence modeling capabilities of LSTMs with the feature extraction capabilities of autoencoders.

Autoencoders, a type of neural network architecture, play a significant role in UAV trajectory prediction by learning compact representations of input trajectories. In the context of

trajectory prediction, Autoencoders aim to encode input trajectories into a lower-dimensional latent space, which typically represents a lower-dimensional space where the model has learned to encode essential features or representations of the input data. Then decode them to reconstruct the original trajectories [105]. The key components and equations for Autoencoders can be expressed as follows:

Autoencoder Architecture:

$$\text{Encoder: } h(\mathbf{x}) = \sigma(\mathbf{W}_1 \cdot \mathbf{x} + \mathbf{b}_1) \quad (23)$$

$$\text{Decoder: } g(\mathbf{h}) = \sigma(\mathbf{W}_2 \cdot \mathbf{h} + \mathbf{b}_2) \quad (24)$$

Here, $\mathbf{x}$ represents the input trajectory, $\mathbf{h}$ is the encoded representation, $\mathbf{W}_1$ and $\mathbf{W}_2$ are weight matrices, $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias vectors, and σ is the activation function.

Objective Function (Reconstruction Loss):

$$\min_{\theta} \sum_{i=1}^n \|\mathbf{x}_i - g(h(\mathbf{x}_i))\|^2 \quad (25)$$

Minimizing the reconstruction loss to ensure that the decoded trajectories closely match the original trajectories.

Training Process:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^n \|\mathbf{x}_i - g(h(\mathbf{x}_i))\|^2 \quad (26)$$

Training the Autoencoder involves finding the optimal parameters θ that minimize the reconstruction loss [170].

Autoencoders provide a powerful way to extract meaningful representations from UAV trajectory data, enabling efficient encoding and decoding processes for trajectory prediction tasks. The encoder maps input trajectory data x to a lower-dimensional latent representation using (23). The decoder reconstructs the trajectories as (24). The model minimizes the reconstruction loss using (25) to ensure decoded trajectories closely match the originals. Training optimizes parameters θ to accurately capture and predict UAV movements. LSTM autoencoders are specifically designed to model several key features. These include the temporal dynamics of the UAV's movement, capturing changes in velocity and acceleration over time. They also encode positional data, representing the UAV's spatial coordinates such as latitude, longitude, and altitude at various time steps. Additionally, the autoencoders can incorporate environmental context by including features like wind speed, weather conditions, and the presence of obstacles, which are critical for accurate trajectory prediction. Furthermore, they can learn behavioral patterns of the UAV, such as recurring flight paths or maneuvers, enhancing the prediction accuracy by understanding the UAV's operational behavior.

Also, autoencoders possess the capability to acquire concise representations of trajectory data, suitable for dimensionality reduction and feature extraction purposes [171]. FloMo [153] is a novel motion prediction model introduced in the context of trajectory prediction for autonomous agents like vehicles and robots. It stands for "Flow-based Motion prediction" and is designed to address the challenges faced by traditional generative models like GANs and VAEs in accurately

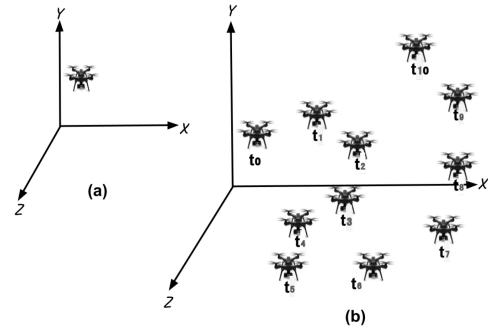


Fig. 12. (a) Shows 3D UAV Space and (b) Shows 4D UAV Space (i.e., Incorporating the Dimension of Time).

predicting future trajectories. A LSTM Auto-Encoder (LSTM-AE) shown in Fig. 11 network-based model has been proposed by Kant et al. [105], which accurately predicts UAV positions in 4D space (shown in Fig. 12) and provides a safe and efficient flight, addressing the challenges of communication and positioning for UAVs. They can capture patterns in sequential trajectory data and generate meaningful latent representations. The study [72] presents an improved multi-step trajectory prediction framework for civil aircraft route planning. It incorporates Bi-directional Long Short-Term Memory Network (Bi-LSTM) and AutoEncoder modules for prediction accuracy. The Bi-LSTM module captures historical trajectory patterns directly, while the AutoEncoder module learns evolutionary features through encoding and decoding. The study [172] addresses predicting the trajectories of other vehicles for safe autonomous vehicle planning. This approach utilizes an LSTM encoder-decoder framework that incorporates spatial attention mechanisms to improve interaction-aware prediction.

F. Variational Autoencoders (VAEs)

VAEs can capture the underlying structure of the trajectory data by learning a probabilistic distribution in the latent space. This allows for generating new trajectories and exploring uncertainties. The study [173] improves agents' predictive abilities by introducing Conditional Flow Variational Autoencoders (CF-VAE) with a novel conditional normalizing flow prior, overcoming limitations of uni-modal Gaussian priors. New regularization techniques enhance training and achieve state-of-the-art results across datasets. The proposed method in [174] is named CSIR, which includes a CS-CVAE (Cascaded Sliding Conditional Variational Autoencoder) module and an I-SCR (Iterative Future-Past Social Compatible Rethinking) module, gives a novel pedestrian trajectory prediction method for applications like surveillance and autonomous driving. CSIR employs cascaded prediction models and an iterative module to address errors, data scarcity, and future-past consistency challenges, outperforming prior methods on benchmark datasets. This study [108] presents a Descriptive Variational Autoencoder (DVAE) for vehicle trajectory prediction, combining neural networks with expert knowledge. The DVAE offers interpretable latent parameters, aiding human understanding and enabling validation using expert-defined rules.

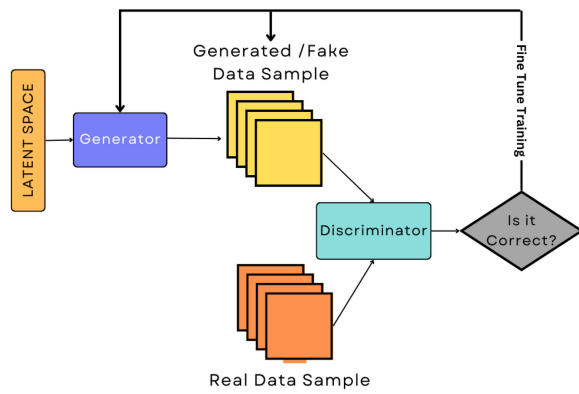


Fig. 13. Working of GANs.

G. Generative AI Methods

Generative AI technologies such as GANs, diffusion models and large language models (LLMs) in the context of trajectory prediction can indeed enhance the depth and relevance of a survey. Given below the discussion on some important GAI models:

1) *Generative Adversarial Networks (GANs)*: GANs can be used to generate new trajectories based on the training data distribution [175], [176]. They help generate diverse trajectory predictions and explore different possible trajectories. GANs consist of a generator and a discriminator. The generator creates data, and the discriminator evaluates its authenticity. They engage in a competitive process, with the generator improving its ability to produce realistic data and the discriminator enhancing its skill in distinguishing real from generated data. This adversarial training continues until the generator creates data indistinguishable from real data [175]. The working of GANs is described in Fig. 13. The study [177] introduces a trajectory prediction model for Unmanned Aircraft Systems (UAS) using a Generative Adversarial Network (GAN). The GAN's generator takes previous trajectory data and produces flight status predictions. This model enhances UAS landing management in busy vertiport airspace by avoiding conflicts. The paper [178] introduces a data-driven algorithm for Aircraft Trajectory Prediction (ATP) using a Neural Network and GAN framework. Blockchain is employed to secure and validate predicted trajectories against adversarial attacks. The study by Wang et al. [12] introduces a novel hybrid 4D trajectory prediction model for short-term predictions in Terminal Maneuvering Area (TMA). The Model integrates clustering and Multi-cell Neural Networks (MCNN), enhancing prediction accuracy; robust preprocessing ensures high-quality data and outperforms the traditional Multiple Linear Regression (MLR) model. Reference [179] proposes a new machine-learning model for classifying drone data in digital forensic investigations, suggesting that well-trained models can accurately analyze drone data from crime scenes, opening possibilities for forensic applications. The paper [180] investigates UAV-operated food delivery services in the absence of real-world data. To address this limitation, an Air Traffic Simulator (ATS) is employed to generate a relevant dataset. Through simulations, the study reveals the enhanced

efficiency of UAVs attributed to their shorter flight paths and the influence of wind conditions on their speeds. The use of air traffic simulators proves instrumental in facilitating realistic simulations of UAV system operations. PECGAN [181] uses a conditional GAN model to predict trajectories by conditioning on predicted endpoints. This method refines motion features using social interactions, for more accurate trajectory prediction that satisfies physical constraints. This study [182] gives a social conditional GAN to predict UAV trajectories, considering social behaviors and environmental constraints. By including observed trajectories and surroundings, the model achieves improved accuracy and generates multiple feasible paths for UAVs, crucial for dynamic and complex environments.

2) *Diffusion Model*: Diffusion models have shown significant potential in various domains of path planning applications due to their ability to generate high-quality data and capture complex distributions [183], [184], [185]. Integrating these models into UAV trajectory prediction could enhance the ability to forecast more accurate and realistic paths, especially in dynamic unpredictable environments. This approach could also enable the generation of diverse trajectory scenarios, providing a robust dataset for training and validating predictive algorithms. The paper [186], address the limitations of existing trajectory prediction methods by proposing an intention-aware denoising diffusion model (IDM) for trajectory prediction. This model is designed to handle the inherent uncertainty of trajectory prediction tasks by decoupling the uncertainty into intention uncertainty and action uncertainty. As this model has been successfully used for trajectory prediction of users, we can also apply it for UAV trajectory prediction to enhance accuracy and efficiency. Additionally, using diffusion models for trajectory prediction could facilitate better handling of multi-modal data, further enhancing the UAVs' situational awareness and decision-making capabilities.

3) *Large Language Models (LLMs)*: LLMs like GPT-4 have demonstrated exceptional capabilities in understanding and generating coherent text. Recently, their use has expanded into areas such as trajectory prediction by using their ability to process large datasets and learn complex patterns. LLMs can be employed to improve the prediction and planning of UAV trajectories by analyzing historical flight data, environmental conditions, and parameters specific to mission, to generate more accurate and adaptive flight paths. For example, an LLM can learn from large amounts of flight data to identify optimal routes and adjust trajectories in real-time based on changing conditions, thus improving the UAVs' safety and efficiency in dynamic environments. LLMs can be pre-trained with datasets comprising various flight scenarios and environmental conditions, allowing them to understand and predict the UAV's movements accurately [187]. This study [188] aims to address how LLMs can optimize communication, adapt to new missions dynamically, and process complex data streams, enhancing UAV efficiency and versatility across various domains, including emergency response, environmental monitoring, urban planning, and satellite communication. The study [187] explores the potential of integrating LLMs into UAV systems to enhance their autonomous flight capabilities.

The paper demonstrates how LLMs can process and analyze data from dynamic obstacles, enabling UAVs to navigate safely and efficiently in real-time. Integrating real-time sensor data allows continuous adaptation to environmental changes, while combining LLMs with other AI models like diffusion models or GANs creates robust hybrid systems. Developing user-friendly interfaces enables natural language interactions for simplified mission planning. When used for onboard processing, LLMs can analyze and process data locally on the UAV, enabling real-time decision-making and reducing latency associated with data transmission to ground stations.

4) *Flow Based Methods*: Flow-based GAI models differ from other models like GANs and VAEs in their explicit density modeling, invertibility, training stability, and application focus. These models are ideal for UAV trajectory prediction application because of their ability to efficiently model and generate data distributions. By training on recorded flight data, these models learn complex trajectory patterns and generate synthetic data that represents real-world paths. This enhances training datasets, optimizes flight paths, and supports real-time adjustments. Integrating flow-based models with path planning algorithms allows for dynamic, efficient, and safe UAV operations in varying environments, ultimately improving resource management and decision-making in UAV-assisted networks [189].

H. Sequence-to-Sequence Models

These models are designed to map input sequences to output sequences. They are well-suited for trajectory prediction tasks where historical trajectory data is used to predict future trajectory sequences. The sequence-to-sequence Recurrent Neural Network in the study [190] is used to predict future vehicle trajectories sequentially, forecasting positions and headings up to 3 seconds ahead with 0.5-second intervals. This model considers the diverse maneuvers at intersections and aims to improve accuracy in trajectory prediction, contributing to more reliable conflict analysis for traffic safety. Sequence-to-sequence methods predict RUL values in [94] for future periods, providing information about the Quadrotor Unmanned Aerial Vehicle (QUAV)'s capability for upcoming maneuvers. In this case, sequence-to-sequence models are employed to accurately forecast the RUL in scenarios where one-step predictions are inadequate.

I. Attention Mechanisms

Attention mechanisms, as used in Transformer models, can be applied to other sequence-based models to focus on specific parts of the trajectory that are relevant for prediction. The attention mechanism is used in the study [120] to facilitate relational reasoning between interactive agents for multi-agent trajectory prediction. It helps capture complex interactions and dependencies between agents in dynamic and crowded scenarios. The attention mechanism, specifically implemented through a Graph Double-attention Network (GDAT), allows the model to focus on relevant agents and their spatiotemporal relationships while predicting trajectories. The paper [109]

introduces an LSTM-based model with an attention mechanism for precise trajectory prediction using a high-precision traffic dataset. The approach accounts for individual driving tendencies, interactions, and road geometry, outperforming previous studies in various scenarios.

Table V given a comparative analysis of Deep Learning based methods on the basis of their Input type, output type, role in UAV trajectory prediction, limitations and their next versions.

J. Limitations of Deep Learning Methods

1) *Data Requirements*: DL models often require large amounts of labeled training data, which might be challenging to collect for UAV trajectory prediction, especially in diverse and complex environments. The limitation of the DL models is that they rely solely on historical trajectory data; the dataset may contain redundant, noisy, or erroneous data, which could impact the accuracy of trajectory predictions. Additionally, the models do not account for dynamic effects or real-time updates, potentially leading to suboptimal predictions in rapidly changing conditions [98]. While Deep Neural Network (DNN) prediction methods have gained popularity for vehicle and pedestrian trajectory prediction, there needs to be more research on applying DNNs to predict UAV trajectories. Previous models focused more on ground-based vehicles and pedestrians [108].

2) *Computational Resources*: A limitation of using deep learning models in UAV trajectory prediction could be the requirement for substantial computational resources; the use of complex neural network architectures, especially for real-time applications like drone racing, might lead to higher processing times, impacting the ability of the drone to make rapid trajectory adjustments [69]. Implementing resource-intensive models on UAVs with constrained hardware could lead to operational challenges and increased energy consumption, affecting the UAV's overall efficiency.

3) *Interpretability*: DL models can be regarded as Opaque systems, rendering it challenging to know the underlying mechanisms by which they generate their predictions, which is crucial for safety-critical applications [135]. A potential limitation of using DL models like Variational Autoencoders (VAEs) for UAV trajectory prediction could be that while they offer interpretable latent spaces and similar prediction accuracy as non-interpretable models, they might still have a constraint in terms of prediction accuracy when compared to more complex baseline VAEs that provide greater degrees of freedom [108]. This limitation could arise from the additional constraints introduced by the interpretability features in the VAE's decoder.

4) *Overfitting*: Deep learning models are prone to overfitting when the training data is limited or noisy, leading to poor generalization to new scenarios [135]. DL models often demand a substantial volume of training data, a task that can present considerable challenges when faced with real-world situations, leading to potential overfitting or generalization issues [69]. In scenarios where the dataset is small or lacks diversity, the deep learning models may struggle to generalize

TABLE V
UAV TRAJECTORY PREDICTION DEEP LEARNING MODELS

Model	Input Data	Output Data	Relationship with Other Models	Role in UAV Trajectory Prediction	Best Suitable Scenario	Limitation	Enhanced Version
RNNs [190], [191]	Sequential Data	Predicted Trajectory	Interacts with Attention Mechanisms	Enhancing Sequence Prediction	Dynamic Environments	Vanishing/Exploding Gradient	LSTM
CNNs [69], [167], [96], [119]	Sensor Data	Spatial Feature Extraction	-	Enhancing Spatial Understanding	Aerial Imaging	Limited to Grid Structures	3D CNNs
Transformers [165], [94]	Sequence Data	Predicted Trajectory	Interacts with Attention Mechanisms	Capturing Long-Range Dependencies	Complex Scenarios	High Computational Cost	Sparse Transformers
LSTMs [98], [98], [118], [109], [108], [28]	Sequential Data	Predicted Trajectory	Interacts with Attention Mechanisms	Improving Long-Term Sequence Prediction	Predicting Flight Paths	Computational Complexity	GRU
LSTM Autoencoders [72], [39], [170]	Sequential Data	Reconstructed Trajectory	Interacts with GANs	Anomaly Detection	Data Anomalies	Sensitive to Noise	Variational Autoencoders
VAEs [192], [173]	Sequential Data	Reconstructed Trajectory	Interacts with GANs	Capturing Latent Features	Uncertain Environments	Mode Collapse	Conditional VAEs
GANs [175]	Random Noise	Generated Trajectory	Interacts with LSTM Autoencoders	Data Augmentation	Limited Training Data	Mode Collapse	Wasserstein GANs
Sequence-to-Sequence Models [191], [110], [94]	Sequential Input	Sequential Output	Interacts with Attention Mechanisms	Sequence Transformation	Dynamic Path Planning	Limited to Fixed-Length Sequences	Transformer Seq2Seq
Attention Mechanisms [110], [109], [172], [119]	Sequence Data	Contextual Output	Interacts with RNNs, Transformers	Improving Information Flow	Adaptive Trajectory Prediction	-	Gated Attention Mechanisms

well to real-world situations, leading to potential overfitting and reduced prediction accuracy [103].

5) *Complexity*: Deep learning models can be very complex, requiring significant computational resources for training and inference, which might not be feasible for onboard processing on UAVs with limited hardware. In UAV trajectory prediction, model complexity could result in longer training times, increased computational requirements, and difficulties in fine-tuning the model for specific UAV operational constraints.

6) *Limited Understanding of Dynamics*: Deep learning models might not fully capture the underlying physics and dynamics of UAV motion, leading to physically unrealistic predictions. While CNNs are excellent at processing spatial features in images, they might need to understand the sequential nature of time-dependent trajectories [167]. In scenarios where drones must predict their future path accurately based on past movements, CNNs might need to pay attention to the nuanced temporal dependencies required for precise trajectory forecasting.

VI. REINFORCEMENT LEARNING MODEL

Reinforcement Learning (RL) can be applied to UAV trajectory prediction tasks to learn optimal policies for making sequential decisions [62], [195]. Table VII gives a comparative analysis of reinforcement learning methods on the basis of their strengths, weaknesses and ideal application scenario. Some popular approaches has been mentioned separately with

details in Table VIII and IX. The following RL methods can be applied to predict UAV trajectories::

A. Deep Q-Network (DQN)

DQN stands as a fundamental RL algorithm, amalgamating the principles of Q-learning with deep neural networks. Its applicability extends to UAV trajectory prediction by considering trajectory design as a sequential decision-making challenge, where the agent (UAV) takes actions (adjusts trajectory) to maximize cumulative rewards (achieving desired trajectory goals).

The Q learning equations works in the given way. The Q-value update equation:

$$Q(s, a) = \mathbb{E} \left[r + \gamma \max_{a'} Q(s', a') \mid s, a \right] \quad (27)$$

The loss function:

$$\mathcal{L}(\theta) = \mathbb{E} \left[\left(Q(s, a; \theta) - \left(r + \gamma \max_{a'} Q(s', a'; \theta^-) \right) \right)^2 \right] \quad (28)$$

The parameter update rule:

$$\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta) \quad (29)$$

Here:

- $Q(s, a)$ represents the Q-value for a state-action pair.

TABLE VI
THE PREVALENT STRATEGIES EMPLOYED USING DEEP LEARNING-BASED METHODOLOGIES

Author	Year	Method	Dataset	Main Findings	Evaluation Metrics	Limitations	Future Scope
Kouris et al. [96]	2018	Two-Stream CNN for Perception	Indoor UAV trajectories is created for training	The distance predictions made by the CNN help inform the local motion planning policy, enabling safe navigation by making informed motion decisions.	Time, RMSE	Limited to indoor environments; requires a pre-trained model for each new environment.	Investigate integration with broader environment awareness and complex dynamics for improved real-world navigation.
Shi et al. [98]	2018	LSTM	ADS-B data from ground stations recorded every 15 seconds, spanning June to November 2017	Proposed LSTM-based method accurately predicts 3-D and 4-D flight trajectories without requiring the physical model of the aircraft.	MAE, MRE, RMSE, DWT	Limited to ADS-B data, noised or redundant, doesn't account for dynamic weather effects, lacks real-time adaptation.	Incorporate multi-modal data (images, audios, videos) and enhance the LSTM-based model to fuse different modalities.
Munay et al. [162]	2019	MLP, LSTM, LDA, K-means	Received Signal Strength (RSS) and Signal-to-Noise Ratio (SNR) measurements	Method for maintaining mobile users within UAV transmission range in dynamic environments while optimizing UAV positioning for maximum throughput.	MSE, RMSE	Lack of noise handling, real-time challenges, potential scalability issues in densely populated scenarios.	Apply DL for indoor-outdoor UAV positioning in urban, suburban, and rural environments.
Cheng et al. [31]	2021	LSTM	ADS-B: July 10, 2019; 510,000 flight, 12:00 PM and 1:00 AM.	It combines trajectory prediction methods and evaluates their impact on conflict prediction accuracy for real conflict situations.	RMSE	Performance degradation in mid-term and long-term predictions impacting conflict detection; reliability enhancement needed.	Enhance model by designing more robust neural networks and integrating pertinent information for better prediction accuracy.
Becker et al. [27]	2021	RNN	Real-world ANTI-UAV dataset	The RNN-based MDN model effectively predicts future positions of objects using the observed image coordinates, enabling trajectory prediction in image space.	Final displacement error (FDE)	Depends on synthetic data, lacks consideration of external factors in real-world scenarios.	Explore incorporating real-world noise and complex environmental factors into trajectory generation.
Kant et al. [105]	2022	LSTM-Autoencoder network.	GPS and weather data: Recorded by multiple Ground Data Terminals (GDTs)	Novel LSTM-AE network for 4D UAV position finding without physical model usage, Mathematical model for calculating Azimuth and Elevation angles for ground radar.	MAE, MSE and RMSE	Limited evaluation with a specific dataset and potential challenges in handling diverse UAV behaviors.	Explore real-time implementation and validation of the LSTM-AE model in practical UAV scenarios.
Zhang et al. [28]	2022	LSTM	ADS-B data, latitude and longitude coordinates (Lat, Lon) are obtained using GPS	Demonstrates promising results in accurately predicting UAV trajectories, which is crucial for tasks such as air traffic control, conflict detection, and emergency obstacle avoidance.	Average Prediction Error	Limited analysis on handling diverse weather conditions and sparse ADS-B data in practical environments.	Investigate RLSTM's performance on more complex scenarios and optimize computation efficiency for real-time applications.
Ortega et al. [69]	2022	CNN	Gazebo simulation, comprises 27,992 grey-scale training images and 5,994 validation images	Inception modules are used in the system to process various perspectives of the same object and extract crucial features for pose learning.	3D pose estimation error	Limited to single-channel images; further optimization for higher efficiency is needed; unaddressed error accumulation.	Enhance GreySeqNet for even faster pose estimation and explore multi-camera configurations for improved accuracy.
Yang et al. [103]	2022	LSTM	Radiation signal of UAV	Creating an LSTM-Chan prediction framework that combines the flight trajectory data of UAVs and LSTM neural network to predict the UAVs' flight trajectories.	MSE	Limited to passive detection, not addressing active radar or sensor data.	Establish a fusion model considering UAV flight path characteristics to enhance accuracy.
Wu et al. [72]	2023	Bi-LSTM-AutoEncoder, voting fusion module.	Flight trajectories: George Bush to Logan International Airports, US	The results of real flight trajectory prediction demonstrate the advantages and effectiveness of the proposed multi-step TP framework.	MAE, RMSE	Dependency on accurate input data, potential sensitivity to noise and errors in real-world flight trajectories.	Investigate the adaptability and robustness of the framework under various weather and environmental conditions.
Zhou et al. [174]	2023	CS-CVAE and I-SCR	ETH/UCY [196](5 scenes, 1536 pedestrians) and SDD [197](8 aerial-view campus scenes with 11000 pedestrians)	The CSIR method demonstrated superior performance compared to other state-of-the-art methods on benchmark datasets for pedestrian trajectory prediction.	MAE, RMSE	Complex interactions in crowded scenes and require fine-tuning for diverse environments.	Exploring multi-modal data integration and adapting the method to real-time dynamic environments.

- r is the immediate reward.
- γ is the discount factor.
- $L(\theta)$ is the loss function.
- θ are the parameters of the Q-network.
- θ^- are the parameters of the target Q-network.
- α is the learning rate.

The UAV's state s includes its current position and velocity, while the action a represents trajectory adjustments. The reward r measures immediate success, such as proximity to a target or energy efficiency. The Q-value $Q(s, a)$ estimates the expected cumulative reward of taking action a in state s . The Q-value update, i.e., (27) helps the UAV evaluate the long-term

TABLE VII
COMPARATIVE ANALYSIS OF REINFORCEMENT LEARNING METHODS FOR TRAJECTORY PREDICTION

Method	Strengths	Weaknesses	Ideal Application Scenarios
Deep Q-Network (DQN)	Efficient with discrete action spaces, suitable for trajectory prediction in scenarios with clear state-action pairs.	Limited to discrete action spaces, may struggle with continuous action spaces in trajectory prediction.	Navigation tasks in grid-world environments, scenarios with well-defined action spaces [40], [196].
Policy-Based Methods			
a. Policy Gradient Methods	Effective in high-dimensional and continuous action spaces. Directly optimizes policy for trajectory prediction.	High variance in gradient estimates. Can be sample-inefficient.	Continuous control tasks in trajectory prediction. Applications with stochastic environments [195].
b. Proximal Policy Optimization (PPO)	Addresses high variance issues. Balances stability and sample efficiency [198], [73].	May have issues with local optima. Complexity in tuning hyperparameters [100].	Robotics applications in trajectory prediction. Tasks requiring a balance between stability and sample efficiency [95].
c. Deep Deterministic Policy Gradient (DDPG)	Suitable for continuous action spaces. Handles deterministic policies in trajectory prediction.	Prone to overestimation bias. Sensitive to hyperparameter tuning.	Robotics tasks with continuous action spaces. Scenarios where deterministic policies are crucial [74], [32], [33].
Actor-Critic Models	Combines benefits of value-based and policy-based methods. Effective in balancing exploration and exploitation [75].	May suffer from high variance. Complexity in training stability [89].	Complex trajectory prediction tasks with both discrete and continuous action spaces. Scenarios where a balance between value estimation and policy optimization is crucial [122], [97].
Model-Based RL	Efficient in sample usage. Suitable for scenarios with limited data [199], [200].	Relies on an accurate model, which may be challenging to learn. Sensitive to model errors.	Trajectory prediction in resource-constrained environments. Applications where real-world data is limited.

benefits of its actions. The loss function mentioned in (28) measures the difference between predicted and target Q-values, and the parameter update rule given in (29) adjusts the Q-network to minimize this difference. This process involves observing the current state, selecting actions, transitioning to new states, and updating the Q-network using experiences stored in a replay buffer. By iteratively refining its policy, the UAV learns to navigate efficiently, avoiding obstacles and optimizing energy use to achieve its trajectory goals. The article [40], introduces a new approach to coverage path planning for UAVs with random start and landing positions, using Double Deep Q-Network (DDQN) to navigate around no-fly zones, considering power constraints. DDQN is an enhancement of the Deep Q-Network (DQN) addressing the overestimation bias present in DQN by decoupling the action selection and action evaluation processes. In UAV trajectory prediction for coverage path planning, DDQN trains a control policy to efficiently guide the UAV while managing power constraints, varying start/landing positions, and no-fly zones. It uses map-like inputs and convolutional networks to optimize coverage and decision-making. The study [196] provides a DRL approach for trajectory optimization and Ground Base Station (GBS) association for cellular-connected UAVs. The objective of the paper [197] is to develop an intelligent UAV trajectory planning method using a Double Deep Q-Network (DDQN) algorithm to optimize data collection in IoT networks while navigating complex urban environments with obstacles and jammers.

B. Policy Based Methods

1) *Policy Gradient Methods*: Policy gradient methods, such as PPO and TRPO which are Proximal Policy Optimization and Trust Region Policy Optimization, respectively, are examples of algorithms that directly optimize the agent's policy. These methods focus on mapping states to actions,

and they can be effectively employed for UAV trajectory prediction. They can learn complex policies for UAVs to navigate in changing environments. An enhanced PPO algorithm in [198] uses the LSTM network to optimize UAV maneuvering based on real-time position and contextual data, achieving efficient and robust autonomous navigation. The challenge of hybrid action space is addressed in [41] using a Probabilistic Multiagent Deep Deterministic Policy Gradient (PMADDPG) approach. AG-PMADDPG is introduced to enhance GU throughput and UAV-BS communication service fairness. Game theory analysis reveals the Price of Anarchy (POA), showing performance loss if GUs learn independently. The study [64] addresses UAV-aided vehicle communication through joint UAV trajectory and vehicle assignment optimization, considering partial observability. A novel Deep Recurrent QN (DRQN) framework is introduced using LSTM to predict vehicle trajectories and optimize them efficiently, outperforming baseline approaches in terms of spectral efficiency. A meta-learning policy gradient algorithm is proposed for optimizing DBS trajectories in dynamic wireless environments with unpredictable user requests; the algorithm outperforms the policy gradient algorithm in terms of speed of convergence, service success rate, and reduction in variance [86].

2) *Proximal Policy Optimization (PPO)*: It is an RL algorithm that aims to optimize an agent's policy while ensuring that policy updates are within a specific range to prevent drastic changes as shown in Fig. 14.

PPO optimizes the policy by iteratively updating parameters θ based on the collected data. The objective is to maximize the expected cumulative reward $J(\theta)$. The PPO objective function involves a surrogate loss, $L(\theta)$, and a clipping mechanism:

$$L(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \cdot \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \cdot \hat{A}_t \right) \right] \quad (30)$$

TABLE VIII
THE PREVALENT STRATEGIES EMPLOYED USING REINFORCEMENT LEARNING-BASED METHODOLOGIES

Author	Year	Method	Dataset	Main Findings	Evaluation Metrics	Limitations	Future Scope
Imanberdiyev et al. [112]	2016	Model-based RL algorithm called TEXPLORE	ROS (Robot Operating System) and Gazebo simulator	For solving real-world challenges in autonomous navigation for UAVs, it learns quickly in few episodes and allows for real-time actions due to its parallel architecture.	Learning efficiency, trajectory quality	The model's performance could vary in complex real-world scenarios, other model learning methods might enhance performance.	Tackling challenges and conducting real-world quadcopter experiments to enhance UAV autonomy.
Liu et al. [35]	2019	Multi-agent learning placement algorithm	Twitter Data; GPS coordinates, time stamps of users	Multi-agent Q-learning and precise user mobility prediction enhance trajectory planning and system efficiency	Learning convergence, Transmit Rate	Limited consideration of real-world uncertainties and dynamic obstacles in trajectory planning and power control.	Advanced ML techniques for more accurate dynamic trajectory adjustments.
Lopes et al. [95]	2018	Proximal Policy Optimization (PPO) algorithm	V-REP simulator and the Vortex physics engine for experiments	Present a quadrotor control approach that uses the PPO algorithm and stochastic policy gradients.	Normalized accumulated reward, Comparison of positions	Stochastic policy might lack precision, limited to simulated environments, and potential instability during certain trajectories.	Explore new reward signals, incorporate classic control theory to reduce steady-state error.
Zhao et al. [100]	2020	Multi-Agent Proximal Policy Optimization (MAJPO)	Simulation using parameter setting in LSTM	MAJPO algorithm's main goal is to learn the joint state-value function and the generalized advantage estimate for evaluating and optimizing distribution policies.	Sum of first-order absolute center moment	The model's effectiveness in highly dynamic and real-world scenarios needs further investigation and testing.	Explore dynamic adaptation of the MAJPO algorithm for more complex and dynamic environments.
Cheng et al. [201]	2020	Soft Actor-Critic (SAC) deep reinforcement learning	Quadrotor drone's dynamic model; plant protection drone parameters	Eliminates steady-state errors, reduces controller saturation, and ensures robustness in controlling the quadrotor's attitude system.	Target angles, divergence issues, and the stability achieved	Algorithm's performance limited by simulated environment vs. real-world disparities. Network downsizing could address computations for on-drone computer.	Exploring distributed training methods for online training in drones to enhance efficiency.
Hu et al. [86]	2020	Meta-learning-based trajectory optimization	Drone base station (DBS) serving 100 mobile users.	Proposed meta-learning approach adapts DBS to changing environments more effectively than previous RL-based and ML-based solutions.	Convergence speed, communication performance, and service coverage probability	Limited to Single DBS, May Not Address All Dynamic Factors	Extend to Multiple DBSs, Incorporate Dynamic Factors
Theile et al. [40]	2020	UAV control method based on double deep Q-learning.	Agent moves in a 2D grid with action commands	The incorporation of map-like channels adds spatial information to convolutional network layers, enhancing the agent's decision-making capabilities.	Coverage ratio, landing success ratio, trajectory planning performance, cumulative reward.	No consideration for scenarios with variable power consumption, Limited to 2D grid environments.	Extend the approach to handle higher dimensions and dynamics, such as altitude and heading.
Ding et al. [41]	2021	Probabilistic multi-agent deep deterministic policy gradient (PMADDPG)	2 km × 2 km square area, UAVs on a 500 m radius circle at 100 m altitude	PMADDPG approach for optimizing UAV-BS trajectories and GU access control in an air-ground coordinated communications system	Sum of the accumulative reward of all agents	Limited to discrete GU actions, oscillation in training, potential boundary constraint violations in early training.	Explore UAV-BS frequency band allocation for improved system performance.
Chen et al. [196]	2021	DRL approach for trajectory optimization	Simulation: 8 GBSS; 4000 m × 4000 m area	Given DRL approach outperforms conventional methods in terms of trajectory length and maintains reliable UAV-GBS connectivity.	Trajectory length, connectivity outage time, communication performance	Limited to rotary-wing UAVs; extension to other types needed. Simplified channel models; real-world variations not fully captured.	Investigate adaptive antenna configurations for UAV-GBS communication.
Hu et al. [83]	2021	Value Decomposition Based Reinforcement Learning (VD-RL)	Simulation: Five DBSS serving 300 mobile users.	VD-RL algorithm enables DBSSs to independently select strategies, achieving higher team utility compared to existing multi-agent RL solutions.	Successful Service Rate	Limited to specific system and environment models, assumptions on user requests, and scalability concerns.	Explore more complex environments, integrate real-world constraints, and enhance scalability for larger networks.

where:

- $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the importance sampling ratio.
- $\hat{A}_t$ is the advantage estimate.
- $\text{clip}(x, a, b)$ clips the value of x to be between a and b .

The update rule for the policy parameters is given by:

$$\theta_{\text{new}} = \arg \max_{\theta} [\mathbb{E}_t[L(\theta)] - c \cdot \text{KL}[\pi_{\theta_{old}}(\cdot|s_t), \pi_{\theta}(\cdot|s_t)]] \quad (31)$$

where:

- c is a hyperparameter controlling the strength of the policy change constraint.
- $\text{KL}[\pi_{\theta_{old}}(\cdot|s_t), \pi_{\theta}(\cdot|s_t)]$ is the Kullback-Leibler divergence between the old and updated policies.

PPO can be used to train UAVs to predict and adjust their trajectories based on rewards obtained from the environment. The proposed method in [101] involves using a

TABLE IX
THE PREVALENT STRATEGIES EMPLOYED USING REINFORCEMENT LEARNING-BASED METHODOLOGIES

Author	Year	Method	Dataset	Main Findings	Evaluation Metrics	Limitations	Future Scope
Wang et al. [102]	2021	TD3 for Trajectory Design for Completion Time Minimization (TDCTM)	Simulation: independent realizations, LoS link establishment, Fixed altitude $H = 95m$	TD3-TDCTM algorithm successfully minimizes mission completion time in UAV-assisted IoT data collection systems.	Successful Service Rate	The proposed TD3-TDCTM assumes merged phenomenon representation and limited information exchange, potentially impacting accuracy in complex environments.	Extend TD3-TDCTM to 3-D trajectory design for multi-antenna UAVs and multi-UAV scenarios, including beamforming and G2A channel estimation.
Deng et al. [64]	2022	Deep recurrent network (DRQN) framework	Simulation: 2 UAVs, altitude: 100 m, airspace: $2 \times 2 \text{ km}^2$ (Time frame = 4 s, Slots = 36).	DRQN framework effectively addresses the POMDP challenge by utilizing only current vehicles' locations for trajectory prediction	Convergence, Performance Comparison	Current model assumes partial observability in vehicle trajectories prediction.	Investigate UAV scheduling considering random small-scale fading for further improvement.
Li et al. [61]	2022	DQN-based algorithm for dynamic UAV trajectory planning	Simulation Based and Field Test	A UAV data collection system on a MEC platform, that adapt to changing conditions in the WSN.	Convergence, Comparisons, Acquisition Success Rate	Partial environmental observations result in uncertain and biased environment models.	Measure environment model uncertainty using belief, improve obstacle avoidance, and include 3D flights.
Liu et al. [74]	2022	Reference Model-based Deep Deterministic Policy Gradient	Simulation environment is built based on pygame, tensorflow	Aims to strike a balance between high return and strong randomness in the strategy. This approach enhances exploration ability and robustness.	Distance between the UAV and the target airport	Lack of detailed analysis on algorithm robustness to uncertainties and hardware limitations.	Explore real-world implementation and testing of the SAC algorithm for UAV autonomy in diverse environments.
Wang et al. [74]	2022	ESB-MADDPG method	Offline trajectory dataset obtained by the expert system	The integration of ESB-MADDPG with expert system-generated offline trajectories improves the efficiency of swarm robot decision-making.	Swarm decision-making efficiency	Offline training time remains a concern; explore faster decision-making methods for swarm robots.	Consider a more general dynamic model for swarm systems beyond wheeled robots.
Hsu et al. [62]	2022	A distributed RL model with convex-TSP algorithm	Real time simulation	A distributed RL approach for UAV collision avoidance and trajectory optimization, efficient goods delivery and data collection.	Trajectory length, distance between UAV trajectory and ground IoT devices	Assumes fixed UAV speeds and simplified communication models.	Implement deep Q-learning to reduce reinforcement learning table size.
Dhuhir et al. [101]	2022	Proximal Policy Optimization (PPO) algorithm	Simulation: UAVs monitor $100m \times 100m$ area using diverse CNNs.	Utilization of the PPO algorithm, to address the NP-hard problem: Trajectory planning and CNN task distribution for UAVs.	Average Cumulative Rewards, learning speed	Formulated problem NP-hard, not suitable for real-time applications, Focus on image classification tasks.	Study interference effects on transmission latency under different communication modes (LoS, NLoS).
Nasr-Azadani et al. [75]	2022	Single-agent and multi-agent Actor-Critic (AC)	UAV's location is determined through the Fast Global K-means (FGKM) scheme	Improved overall network throughput achieved by optimizing UAV initial deployment and 3-D trajectory design.	User access rate	Effective trajectory prediction heavily depends on having access to precise and comprehensive user coordinate data.	Explore time-varying domain segmentation algorithms for multiagent trajectory clustering and resource utilization.
Paravreh et al. [89]	2023	Advanced Continuous Deep Q-Learning.	NS-3 network simulator	Expanding the RL algorithm's action space from discrete to continuous using ACQDL, considering user mobility, proposing a new reward function.	Accumulated data rate of endpoints (users)	Limited to single 6G cell, lacks energy efficiency optimization, and needs validation in real-world scenarios.	Explore energy-aware ACQDL for UAV-B5 deployment, multi-UAV scenarios, vision-based deployment, and 6G inter-cell interference.
Wang et al. [197]	2024	DDQN algorithm for intelligent UAV trajectory planning	Simulated UAV and IoT device positions	DDQN algorithm significantly outperforms traditional Q-learning and DQN algorithms in terms of cumulative average rewards and convergence speed.	Cumulative average rewards, convergence rate, data collection speed	Performance is sensitive to hyperparameter settings, and the approach requires high computational resources.	Encorporating UAV battery limitations, addressing multi-UAV dynamic scenarios, and exploring distributed training

proximal policy optimization (PPO) algorithm to find trajectory paths for UAVs and distribute CNN tasks. The simulation setup involves monitoring a $100m \times 100m$ area using UAVs with different computational capacities. Three CNN networks (LeNet, AlexNet, VGG16) process images captured by UAVs for inference requests. Performance comparisons include QoS coefficients, rewards convergence,

latency, and accuracy among algorithms like PPO, Actor-Critic, greedy heuristic, and random selection. The impact of varying QoS coefficients is examined. Graphs illustrate results and algorithm effectiveness. The paper [73] introduces a reinforcement learning method using PPO to adapt the process noise covariance (Q) in Kalman filter tracking. PPO effectively learns optimal Q values for enhanced tracking

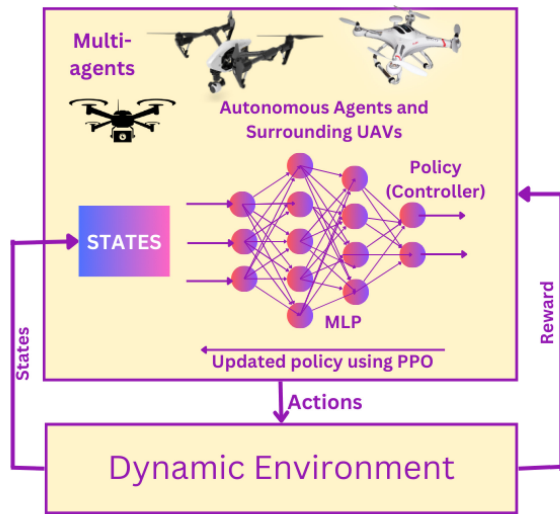


Fig. 14. Proximal-Policy Optimization(PPO) Method.

performance, outperforming other methods, as shown through RMSE comparisons. The work [95] demonstrates the feasibility of utilizing reinforcement learning, specifically the PPO algorithm, to optimize a stochastic control policy for position control of quadrotors. The approach achieves efficient sampling and fast convergence in off-the-shelf simulators without additional exploration strategies. The Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm and Deep Deterministic Policy Gradient (DDPG) algorithm are applied in [33] to train the UAV movement model using situational information, the TD3 algorithm exhibits superior performance in decision-making and efficiency compared to DDPG, rendering it well-suited for combat scenarios. This method empowers UAVs to autonomously execute intelligent maneuvering decisions, using data such as position, speed, and relative azimuth information during air combat. The MultiAgent Joint Proximal Policy Optimization (MAJPPO) algorithm in [100] integrates centralized learning and decentralized execution, enhancing cooperation and rewards in dynamic scenarios, such as multi-UAV formation and navigating obstacles.

3) *Deep Deterministic Policy Gradient (DDPG)*: DDPG can be used for UAV trajectory prediction in scenarios where the trajectory adjustments are continuous and require precise control. The study [102] explores a 3D UAV-assisted IoT system where UAV optimizes its trajectory to collect data from IoT nodes. Imperfect CSI in urban environments is considered, aiming to minimize data collection time using TD3 reinforcement learning. Simulation results confirm its effectiveness against non-learning-based baselines. This study [74] introduces a control strategy for quadrotor drones using deep reinforcement learning. Unlike conventional methods, a deep neural network is directly employed to map drone position and target values to control commands, addressing oscillations and instability issues. The proposed approach combines deep deterministic policy gradient with a reference model for improved robustness and stability. Experimental validation via simulations and actual drone experiments

showcases its effectiveness in achieving stable and accurate control, eliminating oscillations and steady errors. The article [32] introduces an Expert System-based Multiagent Deep Deterministic Policy Gradient (ESB-MADDPG) approach for swarm robot decision-making. ESB-MADDPG enhances training speed, utilizes smooth trajectories, and integrates expert system-generated offline trajectories for efficient swarm robot decision-making using Model Predictive Control (MPC). The article [41] studies UAV-assisted communications with coordinated trajectories of UAV-BSs and Ground User (GU) access control. A cooperative-competitive game is formulated to optimize throughput and fairness, solved using a multiagent deep reinforcement learning approach, AG-PMADDPG, demonstrating improved simulation results.

C. Actor-Critic Models

Actor-critic methods combine value-based and policy-based approaches in reinforcement learning. The actor selects actions based on policy, while the critic evaluates action values. Variants like Asynchronous Advantage Actor-Critic (A3C) and Advantage Actor-Critic (A2C) are effective for UAV trajectory prediction. The paper [83] addresses trajectory design for energy-constrained drones serving ground users with varying demands. It proposes a Value Decomposition Based Reinforcement Learning (VD-RL) approach, ensuring coverage and optimizing convergence speed. The paper [89] presents a continuous actor-critic DRL solution for the dynamic positioning of UAV-BSs considering mobile endpoints. Simulation results show superiority over Q-learning, Deep Q-learning, and conventional algorithms, achieving higher data rates and faster optimization. The paper [201] employs the Soft Actor-Critic (SAC) deep reinforcement learning algorithm to tackle autonomous decision-making for Unmanned Aerial Vehicles (UAVs). It abstracts the UAV decision-making as a game scenario, builds a simulation environment, models the problem as a Markov Decision Process (MDP), and connects SAC with the environment. The study [75] addresses a wireless network with UAVs as aerial base stations and ground users in urban areas, which aims to enhance downlink rates via clustering, accurate user prediction with deep ESN, and efficient UAV deployment using single- and multiagent AC algorithms, improving performance compared to value-based methods. The paper [97] addresses the challenge of autonomous unmanned Aerial Vehicle Navigation (AUN) using DRL. Additionally, an unsupervised classification method is integrated to enhance the reward function.

D. Model-Based RL

Instead of directly learning a policy, model-based RL involves learning a model of the environment dynamics and then planning trajectories based on the learned model. This approach can be helpful for UAV trajectory prediction when simulating different trajectory scenarios before execution. The work [199] employs model-based reinforcement learning with a learned probabilistic drone dynamics model for thrust-attitude control, achieving effective drone control with minimal real-world interactions and onboard resources.

The results demonstrate successful drone control using only onboard resources and sensors, showcasing efficient learning with limited experience. The paper [112] introduces a novel model-based RL algorithm called TEXPLORE for autonomous navigation of UAVs. TEXPLORE adapts to changing conditions and learns efficient trajectories, outperforming Q-learning methods in real-time UAV experiments.

The proposed model-based approach in [200] combines reinforcement learning with Bayesian MPC to efficiently implement policies, ensuring data-efficient learning through posterior sampling. The algorithm's performance is analyzed by establishing reward function continuity and bounding unsafe learning episodes.

The paper by Liu et al. [35] introduces a new framework for the 3D trajectory designing of UAVs, formulating a throughput maximization problem. It presents a three-step approach involving multiagent Q-learning and echo state network for mobility prediction, enabling cooperative trajectory design and power control with demonstrated convergence. The framework presented in the cited paper [61] introduces a two-level DRL approach to tackle the challenge of UAV trajectory planning in dynamic scenarios where the positions of nodes are not fixed. The first-level Deep Neural Network (DNN) models the changing environment, enabling real-time updates for trajectory planning.

E. Limitations Of Reinforcement Learning Methods

Given below are the limitations of using reinforcement learning algorithm in UAV trajectory prediction:

1) *Sample Efficiency*: Reinforcement learning often requires a large number of interactions with the environment to learn effective policies, which can be time-consuming and impractical for real-time UAV trajectory prediction. High computational complexity and resource requirements during training, limiting real-time decision-making in certain scenarios and reliance on large amounts of training data, making the algorithm sensitive to data quality and distribution [33].

2) *Generalization*: Reinforcement learning models might struggle to generalize to new or unseen environments, leading to poor performance in scenarios that differ from the training environment. Existing RL-based solutions are over-fitted to specific environments and are not suitable for rapidly adapting to new or dynamic scenarios due to manually pre-determined internal dynamics (hyper-parameters, exploration strategies, loss functions) [39].

3) *Stability and Convergence*: Reinforcement learning algorithms can be sensitive to hyperparameters, making them prone to instability and slow convergence if not tuned properly. Previous approaches like artificial rule algorithms and traditional methods lack adaptability and self-learning. While deep reinforcement learning (DRL) has shown promise, existing research has limitations such as two-dimensional simulation environments and the lack of consideration for radar and weapon impact [33]. Training the DRQN models may involve complex computation and optimization processes [64].

4) *Partial Observability*: UAV trajectory prediction often involves dealing with partial observations, where not all relevant information is available, which can make policy learning more challenging. RL models often require a significant number of interactions with the environment to learn effective policies, which might not be feasible in fast-paced scenarios such as UAV racing. Rapidly changing conditions, unpredictable actions of other agents, and the need for quick decision-making can hinder RL models from providing accurate and timely trajectory predictions [100].

5) *Curse of Dimensionality*: In situations involving high-dimensional state or action spaces, the search for optimal policies becomes more challenging, and the learning process can become inefficient. A limitation of Q-learning is its struggle with scalability to handle large state and action spaces effectively. In scenarios with numerous possible states and actions, the Q-learning algorithm can become computationally expensive and may require significant time and resources to converge to optimal policies. This is known as the "curse of dimensionality," which can hinder the practical applicability of Q-learning in complex and high-dimensional environments [75].

6) *Transferability*: Transferring learned policies from simulation to real-world UAVs might be difficult due to differences in dynamics, sensor noise, and environmental conditions [202]. An RL model trained in one environment may struggle to adapt and perform optimally in a different setting, potentially compromising safety and mission success. Addressing these challenges includes techniques like domain adaptation, transfer learning, and realistic simulations to enhance model generalization [202], [203].

VII. EVALUATION METRICS

Here is a list of evaluation metrics for assessing the performance of UAV trajectory prediction models, encompassing traditional mathematical models, ML techniques, DL approaches, RL methods, and deep reinforcement learning (DRL) models has been given below and Table X gives a list of these metrics with their category, best scenario in which they can be used some considerations which must be followed for their implementation.

A. Mean Squared Error (MSE)

The Mean Square Error (MSE) is a widely employed metric for assessing the precision of an estimator or prediction. It quantifies the average squared disparity between the predicted and actual values [204]. The formula for MSE is given by:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (32)$$

where:

- n denotes the number of samples,
- y_i denotes the true value of the i -th sample,
- $\hat{y}_i$ denotes the estimated value of the i -th sample.

TABLE X
EVALUATION METRICS FOR UAV TRAJECTORY PREDICTION

Metric	Category	Practical Applicability	Considerations
MSE	Regression	Overall accuracy assessment	Sensitivity to outliers
MAE	Regression	Average magnitude of errors	Less sensitive to outliers
RMSE	Regression	Interpretable metric	Sensitive to outliers
MAPE	Regression	Percentage accuracy	Sensitivity to zero/near-zero actual values
R2 coefficient	Regression	Model fit indication	Scaled between 0 and 1
Precision, Recall, F1-Score	Classification	False positive/negative trade-off	Precision and recall trade-off
AUC	Classification	Distinguishing between classes	Imbalanced dataset considerations
Mean IoU	Reinforcement Learning	Spatial accuracy	Semantic segmentation relevance
Success Rate	Reinforcement Learning	Overall success assessment	Application-specific success criteria
Average Reward per Episode	Reinforcement Learning	Agent's reward accumulation	Reinforcement learning tasks
Convergence Rate	Reinforcement Learning	Learning algorithm speed	Real-time applications
Exploration Efficiency	Reinforcement Learning	Exploration and exploitation balance	Dynamic environments
Learning Speed	Reinforcement Learning	Knowledge acquisition speed	Time-sensitive applications
Policy Entropy	Reinforcement Learning	Decision-making uncertainty	Exploration and exploitation balance
Gini Impurity	Reinforcement Learning	Distribution inequality	Balanced predictions

B. Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) serves as a frequently utilized metric for evaluating the accuracy of an estimator or prediction. It calculates the average absolute deviation between the estimated and actual values [205]. The formula for MAE is given by:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (33)$$

where:

- n denotes the number of samples,
- y_i denotes the true value of the i -th sample,
- $\hat{y}_i$ denotes the estimated or predicted value of the i -th sample.

The MAE metric provides insight into the average magnitude of errors in the predictions, without considering the direction of the errors. It exhibits lower sensitivity to outliers compared to the Mean Square Error (MSE) since it does not involve squaring.

C. Root Mean Squared Error (RMSE)

The Root Mean Square Error (RMSE) is a common metric used to measure the average magnitude of the differences between predicted ($\hat{y}_i$) and actual (y_i) values. It assesses the accuracy of a prediction model [204]. The RMSE formula is given by:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (34)$$

where:

- n is the number of data points.
- y_i represents the actual observed value.
- $\hat{y}_i$ represents the predicted or estimated value.
- The summation goes from $i = 1$ to n , covering all data points.

The square root of MSE estimates the average error magnitude. It helps interpret the scale of errors in the prediction. Studies like [31] have utilized RMSE to evaluate the model.

D. Mean Absolute Percentage Error (MAPE)

MAPE, which stands for Mean Absolute Percentage Error, is a widely employed metric for assessing the accuracy of predictions or estimates, particularly in domains like forecasting and time series analysis [206]. It measures the average percentage difference between the estimated and predicted values. The formula for MAPE is as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (35)$$

where:

- n denotes the number of samples,
- y_i denotes the true value of the i -th sample,
- $\hat{y}_i$ denotes the estimated or predicted value of the i -th sample.

The MAPE metric provides a relative measure of the accuracy of predictions, which is particularly useful when dealing with data on different scales. However, it is essential to note that MAPE can be sensitive to zero values in the true values ($y_i = 0$), which may result in undefined or infinite values. In such cases, modifications to the formula or alternative metrics may be necessary.

E. R-Squared (R2) Coefficient

The R-squared (R2) coefficient, often referred to as the coefficient of determination, is a statistical gauge that signifies the fraction of variance in the dependent variable clarified by the independent variables within a regression model [207]. This metric is frequently utilized to evaluate the goodness of fit of a regression model.

The formula for calculating the R-squared coefficient is given by:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (36)$$

where:

- n denotes the number of observations,
- y_i denotes the observed value of the dependent variable for the i -th observation,
- $\hat{y}_i$ denotes the predicted value of the dependent variable for the i -th observation based on the regression model,
- $\bar{y}$ denotes the mean of the observed values.

The R-squared coefficient ranges from 0 to 1. A higher R-squared value suggests that the independent variables explain more of the variance in the dependent variable. However, a high R-squared doesn't guarantee a good fit, as overfitting is possible. It's crucial to examine other metrics and assess the model's assumptions and residuals.

F. Precision, Recall, F1-Score

Precision, Recall, and F1 Score are commonly used metrics for evaluating the performance of classification models, particularly in binary classification tasks [114].

1) *Precision*: Precision is a metric that assesses the ratio of true positive predictions to all positive predictions generated by the model. It is computed using the following formula:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

2) *Recall*: Recall, also referred to as Sensitivity or True Positive Rate, gauges the ratio of true positive predictions to all actual positive instances within the dataset. It is determined using the following formula:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

3) *F1 Score*: The F1 Score represents the harmonic mean of Precision and Recall, delivering a balanced assessment of the model's performance. It is determined using the following formula:

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Precision-Recall trade-off is a common consideration in classification models. High Precision implies a low rate of false positive predictions, whereas high Recall indicates a low rate of false negative predictions. The F1 Score, combining both metrics, offers a balanced single value. These metrics are employed in classification tasks to assess a model's capacity to correctly classify positive and negative instances:

Precision: The ratio of true positive predictions to all positive predictions.

Recall: The ratio of true positive predictions to actual positive instances.

F1-Score: A harmonic mean of precision and recall.

G. Area Under the Curve (AUC)

The Area Under the Curve (AUC) stands as a widely adopted metric for assessing the effectiveness of binary classification models [208]. It quantifies the model's capability to differentiate between positive and negative classes by considering various threshold values.

1) *AUC Calculation*: The AUC is calculated by integrating the Receiver Operating Characteristic (ROC) curve. The ROC curve represents the True Positive Rate (Recall) against the False Positive Rate at various threshold settings.

$$\text{AUC} = \int_0^1 \text{ROC Curve } d\text{FPR}$$

Here, ROC Curve represents the function that maps False Positive Rate (FPR) to True Positive Rate (Recall).

2) *Interpretation*: The AUC value spans from 0 to 1, with a higher value signifying a superior model performance. An AUC of 0.5 suggests that the model performs at random chance levels, whereas an AUC of 1 signifies flawless classification.

H. Mean IoU (Intersection Over Union)

The Mean Intersection over Union (Mean IoU) is a frequently employed metric for assessing the efficacy of semantic segmentation models. It evaluates the quality of object instance segmentation by computing the ratio of the intersection to the union between the predicted and ground truth regions [209].

1) *IoU Calculation*: Given a set of classes, let TP_k be the true positive instances of class k , FP_k be the false positive instances of class k , and FN_k be the false negative instances of class k . The IoU for each class k is calculated as:

$$\text{IoU}_k = \frac{TP_k}{TP_k + FP_k + FN_k}$$

The Mean IoU is then computed as the average IoU over all classes:

$$\text{Mean IoU} = \frac{1}{N_{\text{classes}}} \sum_{k=1}^{N_{\text{classes}}} \text{IoU}_k$$

2) *Interpretation*: The Mean IoU metric provides a measure of how well a segmentation model is able to accurately delineate object boundaries. Higher Mean IoU values indicate better segmentation performance, where a value of 1 represents perfect segmentation alignment with the ground truth.

I. Success Rate

Given a set of test samples, the Success Rate is calculated by determining the percentage of correctly localized objects based on a predefined threshold of overlap. Let N_{total} be the total number of test samples, and $N_{\text{successful}}$ be the number of samples where the overlap between the predicted bounding box and the ground truth bounding box exceeds the threshold.

The Success Rate is given by:

$$\text{Success Rate} = \frac{N_{\text{successful}}}{N_{\text{total}}} \times 100\%$$

It is commonly used in RL tasks to assess the rate at which the model achieves successful outcomes.

J. Average Reward Per Episode

The Average Reward per Episode is a fundamental metric used to evaluate the performance of reinforcement learning algorithms [210]. It quantifies the average cumulative reward an agent receives over a series of episodes during training.

In a reinforcement learning setting, an agent interacts with an environment by actions and receiving rewards. Let R_t represent the reward obtained by the agent at time step t in an episode, and T denote the total number of time steps in an episode. The Average Reward per Episode (AR) is calculated by summing up the rewards obtained in each episode and dividing by the total number of episodes N :

$$AR = \frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t=1}^{T_i} R_t^{(i)}$$

Here, T_i is the number of time steps in episode i , and $R_t^{(i)}$ is the reward obtained by the agent at time step t in episode i .

K. Convergence Rate

The convergence rate is used to measure how quickly an iterative optimization algorithm approaches its optimal solution. It quantifies the speed at which the algorithm converges to the desired outcome [211]. The convergence rate metric is typically expressed as:

$$\text{Convergence Rate} = \frac{\log(e_{k+1}) - \log(e_k)}{\log(e_k) - \log(e_{k-1})}$$

where:

- e_k represents the error or objective function value at iteration k .
- The logarithmic nature of the formula allows for assessing how fast the error decreases across consecutive iterations.

This metric helps determine the algorithm's efficiency in reaching a solution and can guide the choice of optimization methods.

Measures how quickly an RL or DRL model converges to optimal behavior as shown in [62].

L. Exploration Efficiency

Exploration Efficiency is a metric used to assess the effectiveness of an exploration strategy employed by a reinforcement learning agent. It quantifies how efficiently the agent explores its environment to learn about different states and actions [195].

In a reinforcement learning scenario, an agent navigates the environment to uncover the optimal policy for maximizing cumulative rewards. Exploration Efficiency (EE) quantifies the agent's exploration effectiveness by assessing the ratio of unique states visited to the total number of states within the environment:

$$EE = \frac{\text{Number of Unique States Visited}}{\text{Total Number of States}}$$

The higher the Exploration Efficiency, the more effectively the agent explores the state space. Assesses how efficiently an RL or DRL model explores the action space.

M. Learning Speed

Learning Speed is a metric used to quantify the rate at which an agent learns and improves its performance over time in a reinforcement learning task [212].

In reinforcement learning, the Learning Speed (LS) can be calculated by measuring the rate of change in the agent's performance metric (such as the cumulative reward or accuracy) over a specific period:

$$LS = \frac{\text{Change in Performance Metric}}{\text{Time Interval}}$$

where the change in the performance metric can be computed as the difference between the final and initial values of the metric, and the time interval represents the duration over which the change occurred. Measures how fast an RL or DRL model learns optimal policies.

N. Policy Entropy

Policy Entropy is a metric used to quantify the level of uncertainty or randomness in the actions chosen by an agent's policy in reinforcement learning [213]. In reinforcement learning, the Policy Entropy (H) is calculated based on the probabilities of actions ($p(a)$) chosen by the agent's policy:

$$H(\pi) = - \sum_a p(a) \log p(a)$$

where a represents the set of possible actions, $p(a)$ is the probability of choosing action a according to the policy π , and $\log$ is the natural logarithm.

O. Gini Impurity

Gini Impurity is a measure commonly used in decision tree algorithms to evaluate the impurity or disorder of a set of data samples based on their class distribution [214].

Given a set S with K classes, let p_k be the probability of a randomly selected sample belonging to class k . The Gini Impurity (I_G) for set S is calculated as follows:

$$I_G(S) = 1 - \sum_{k=1}^K (p_k)^2$$

where k ranges over all the classes.

The metrics for UAV trajectory prediction offer valuable insights into UAV performance. Metrics like MSE, MAE, and RMSE are crucial for assessing the accuracy of predicted flight paths, with MSE and RMSE highlighting significant deviations and MAE providing a balanced error magnitude assessment. MAPE and R2 coefficients offer percentage-based accuracy and model fit evaluation, respectively, which are vital for understanding prediction reliability. Classification metrics such as precision, recall, F1-score, and AUC are vital for classifying flight conditions and detecting anomalies. In reinforcement learning, metrics like mean IoU and success rate measure spatial accuracy and mission success. Average reward, convergence rate, exploration efficiency, learning speed, policy entropy, and Gini impurity provide insights into learning effectiveness and decision-making. Together, these metrics help refine algorithms, improve safety, and enhance UAV performance.

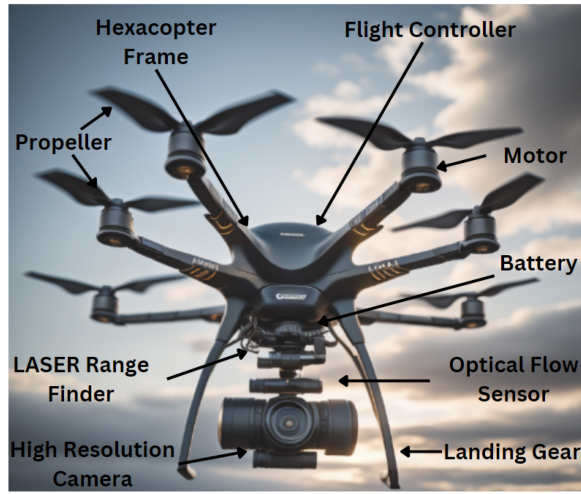


Fig. 15. UAV: Equipped with different Sensors.

VIII. DATASET

UAVs rely on a diverse array of sensors to enable accurate trajectory prediction and ensure safe and precise flight operations. Fig. 15 shows an image of UAV with different sensors. In this section we present a list of sensors which can be used to get the data from UAVs, which can be used for the optimal trajectory prediction. To construct a dataset suitable for such analysis, a multifaceted approach is essential. Data can be sourced from real-world UAV missions equipped with diverse sensors, including GPS, IMU, Lidar, Camera, and Radar, capturing data during missions across different sectors and scenarios. Let us have an example of the GPS (Global Positioning System) sensor for UAV trajectory prediction:

Sensor: GPS (Global Positioning System)

Installation: The GPS sensor is typically mounted on the UAV's frame or fuselage in an unobstructed view of the sky to ensure optimal signal reception. It communicates with satellites to determine the UAV's precise location.

Data Collection: The GPS sensor continuously receives signals from multiple satellites, calculating the UAV's latitude, longitude, altitude, and sometimes velocity. The collected data is timestamped, providing a temporal dimension.

Features of Data:

- 1) *Latitude and Longitude:* Geographic coordinates indicating the UAV's position.
- 2) *Altitude:* The UAV's height above a reference point.
- 3) *Velocity:* Speed and direction of UAV movement.
- 4) *Timestamp:* The time at which each position fix is obtained.

Equations

- 1) *Haversine Formula:*

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R \cdot c$$

where ϕ is latitude, λ is longitude, R is the Earth's radius, and Δ represents differences.

Use for UAV Trajectory Prediction:

- 1) *Positional Tracking:* GPS data enables tracking the UAV's spatial trajectory.
- 2) *Waypoint Navigation:* Sequential GPS coordinates form waypoints for planned routes.
- 3) *Velocity Prediction:* Changes in GPS coordinates over time offer insights into UAV speed and direction.

The GPS sensor, with its accurate positioning capabilities, plays a crucial role in UAV trajectory prediction, forming the basis for navigation, route planning, and overall flight path prediction.

It is crucial to ensure diversity in the dataset, encompassing urban areas, rural landscapes, coastal regions, disaster-stricken areas, and other relevant domains. The dataset should also account for dynamic conditions, such as changing weather and varying light conditions, to enhance its adaptability. To simulate human-in-the-loop scenarios, data from situations where UAVs are operated with human control can be incorporated.

Key statistics for the dataset should include a range of trajectory lengths to address short-term and long-term prediction scenarios. The dataset should consist of an ample number of trajectories to provide a robust dataset for training and evaluation.

Documenting the resolution and quality of sensor data, as well as capturing data at various temporal resolutions, is crucial to informing researchers about the level of detail available. In terms of the train-test split, a random split is recommended to ensure an unbiased representation of data in both the training and testing sets. Optionally, a stratified split can be considered if datasets have specific characteristics that need to be maintained in both sets. It is essential to set aside a validation set for hyper-parameter tuning and model selection during training.

The dataset should cover a variety of domains and scenarios, including urban environments for city navigation and surveillance, rural landscapes for agricultural monitoring and wildlife conservation, coastal regions for maritime surveillance, and disaster-stricken areas for scenarios related to disaster response and recovery. Additionally, the dataset should highlight scenarios where UAVs are operated both autonomously and with human control, reflecting different levels of autonomy. Further we have included a list of sensors and their main properties which can assist the trajectory prediction process and also for more detailed information of the sensors we have included Table XI.

A. GPS (Global Positioning System)

GPS provides accurate information about the drone's latitude, longitude, and altitude. Essential for navigation, positioning, and route planning [215]. GPS stands out as a cornerstone, offering real-time latitude, longitude, altitude, speed, and time data, which are crucial for determining the UAV's precise location and velocity.

TABLE XI
UAV SENSOR CHARACTERISTICS

Name	Use	Output Data	Precision	Availability	Suitable Scenario	Cost	Requirements	Limitations
GPS	Positioning	Coordinates	High	Global	Various	Moderate	Open Sky	Signal Blockage
IMU	Attitude Measurement	Acceleration, Angular Rates	High	Common	All Environments	Low-Moderate	-	Drift Over Time
ADS-B	Surveillance	Aircraft Information	High	Common	Air Traffic Areas	Moderate	ADS-B Network	Limited in Remote Areas
Barometer	Altitude Measurement	Atmospheric Pressure	Moderate	Common	Altitude Control	Low	-	Affected by Weather Changes
Compass	Orientation	Magnetic Field Data	Moderate	Common	Navigation	Low	Calibrated Environment	Susceptible to Interference
Lidar	Obstacle Detection	3D Point Cloud	High	Common	Obstacle-rich Environments	High	Power Supply	Limited Range in Adverse Conditions
Ultrasonic Sensors	Altitude Measurement	Distance to Objects	Low-Moderate	Common	Close-range Navigation	Low	-	Limited Accuracy
Monocular Camera	Imaging	2D Images	Moderate	Common	Visual Navigation	Moderate	Processing Power	Dependent on Lighting Conditions
GNSS	Positioning	Coordinates	High	Global	Various	Moderate	Open Sky	Signal Blockage
Airspeed Sensor	Airspeed Measurement	Airspeed	Moderate	Common	Airspeed Control	Moderate	-	Affected by Winds
Gyroscope	Angular Velocity Measurement	Angular Velocity	High	Common	Attitude Control	Low-Moderate	-	Drift Over Time
Temperature and Humidity Sensors	Environmental Monitoring	Temperature, Humidity	Moderate	Common	Environmental Sensing	Low	-	Limited Range
Multispectral Sensors	Imaging	Multispectral Images	Moderate	Limited	Precision Agriculture	High	Processing Power	Limited to Specific Applications
RF Sensors	Communication	Radio Signals	High	Limited	Signal Intelligence	High	Processing Power	Limited Range

B. Automatic Dependent Surveillance-Broadcast

Surveillance technology used in aviation that enables aircraft to determine their position via satellite navigation and periodically broadcast it, along with other information such as altitude, velocity, and identity, to air traffic control ground stations and other aircraft in the vicinity [216]. It enhances situational awareness and aids air traffic management, allowing for more accurate aircraft tracking and collision avoidance. These dataset types have been widely used in many studies like [31].

C. Inertial Measurement Unit (IMU)

It combines accelerometers, gyroscopes, and, at times, magnetometers to capture a drone's acceleration, angular velocity, and orientation [217]. They are used for stabilization, attitude control, and estimating a drone's movement. IMU complements GPS by providing acceleration and gyroscope readings to estimate the UAV's attitude and velocity changes [218].

D. Barometer

A barometer measures atmospheric pressure to estimate the UAV's altitude. It helps in determining the drone's vertical position and altitude [219].

E. Compass/Magnetometer

Measures the drone's orientation concerning the Earth's magnetic field [220]. Assists in determining heading and direction. A compass or Magnetometer provides heading information, bolstering navigation accuracy, though it can be sensitive to local magnetic disturbances [221].

F. Lidar (Light Detection and Ranging)

It uses laser pulses to create a 3D map of the environment. Provides accurate distance measurements to detect obstacles and terrain. For obstacle detection and terrain mapping, Lidar sensors generate detailed 3D maps [222], [223].

G. Ultrasonic Sensors

Measures distance using sound waves' time-of-flight. They are used for altitude control and obstacle avoidance at close ranges. Ultrasonic sensors measure ground distance, which is particularly useful for low-altitude operations [125], [224], [225].

H. Monocular Camera

Captures 2D images of the surroundings. They are used for visual navigation, object detection, and computer vision

tasks. Visual sensors, such as Monocular Cameras and RGB-D Sensors, capture 2D images or 3D data with depth information, aiding in visual-based navigation and obstacle avoidance [226], [227].

I. GNSS (Global Navigation Satellite System)

It includes systems like GPS, GLONASS, Galileo, and BeiDou. Provides accurate global positioning information for navigation [228]. GNSS offers increased accuracy in tandem with GPS.

J. Airspeed Sensor

It measures the drone's airspeed relative to the air. It helps in controlling the drone's speed and stabilizing flight. Airspeed Sensor assists in maintaining stable flight, especially in varying wind conditions [131], [229], [230].

K. Gyroscope

It measures the angular velocity or rotation rate of the drone and aids in stabilizing and controlling the drone's orientation. Gyroscopes provide crucial data for attitude control [231].

L. Temperature and Humidity Sensors

Monitors atmospheric conditions that can affect drone performance. It helps adjust flight parameters based on environmental factors. Temperature and Humidity Sensors help estimate air density and its impact on UAV performance [232], [233].

M. Multispectral Sensors

Multispectral Sensors capture data in multiple spectral bands (e.g., infrared, ultraviolet) [234]. They are used in agriculture for analyzing crop health and other specialized applications [235].

N. Radio Frequency (RF) Sensors

It measures signal strength and interference in the radio frequency spectrum [236]. It helps assess communication quality and potential interference. Multispectral Sensors go beyond the visible spectrum, enabling specialized applications like crop health assessment [237].

The fusion of data from these sensors equips UAVs with the intelligence needed to accurately predict trajectories and ensure safe operations in various conditions and environments, making them versatile tools for various applications. Table XII shows a detailed list of datasets used in different studies for the evaluation of the models.

IX. SUMMARY

Table XIII shows the summary which presents the findings of a comprehensive survey conducted to analyze and compare the performance of various modeling approaches, specifically focusing on Mathematical Models, ML, DL and RL models. The goal was to offer insights into the respective strengths

and weaknesses of these modeling paradigms across various application domains.

Computational efficiency is also a critical factor in evaluating trajectory prediction methods for UAV networks, especially considering the limited onboard processing capabilities of UAVs. The computational efficiency of these methods can be assessed through three important metrics: algorithmic complexity, processing time, and resource consumption. Algorithmic complexity, expressed in Big-O notation, indicates how the computational requirements with number of operations. Processing time refers to the duration required for the method to execute, which is particularly important for real-time applications. Resource consumption involves the computational resources such as CPU usage, memory, and energy required to run the method. We analyzed the computational efficiency of various trajectory prediction methods and summarized our findings in Table XIV, which provides a detailed comparison of the methods from the latest and most relevant studies for a comprehensive overview, highlighting their respective algorithmic complexities, processing times, and resource consumption, thereby offering a comprehensive overview of their computational demands.

In [26] number of time steps is represented as T , d is the dimensionality of the input and s is the size of the hidden state. For [102] In Training Phase, n represents the number of training iterations required for the TD3 algorithm to converge to an optimal policy. In Execution Phase, n represents the number of real-time steps the UAV takes during the execution of the pre-trained policy. Similarly for [28] T is the number of time steps, d is the input dimension, and h is the hidden layer dimension. The LSTM Auto-Encoder [105] has a linear algorithmic complexity of $O(n)$, which means the time it takes to complete the computation increases linearly with the size of the input data. In [130] both the given methods employ quasi-convex optimization solved using the successive convex approximation (SCA) technique. In the complexity given in Table XIV, A is the number of SCA iterations, $c(K)$ is the complexity of solving the convex feasibility problem, with K being the number of optimization variables and ϵ represents the solution accuracy. The study [41] aims to enhance flight safety and conflict detection by integrating trajectory prediction methods where T is the length of the input sequence. The optimization problem is formulated as a mixed integer non-linear programming (MINLP) problem, which is proven to be NP-hard with no solution in polynomial time. In [73] $O(T)$, where T is the number of timesteps in the episode. The complexity is due to the iterative update process of the policy. In [197] S is the state space size, A is the action space size, and N is the number of neurons in the hidden layers.

Different methods provide these critical onboard processing capabilities to varying extents. Reinforcement learning methods like TD3 [102] and ACDQL [89], while initially computationally intensive, become efficient post-training, making them suitable for deployment where real-time execution is essential. These methods utilize onboard GPUs to manage energy and buffer size effectively. PPO [73] and DDQN [197] manage to balance performance and resource

TABLE XII
DATASET USED IN VARIOUS STUDIES

Reference	Year	Source	Description	Time Period	Size	Area/ Coverage	Data Type/ Features
[150]	2016	Real host track data for Delta Airlines' flights DAL2173	Empirical evaluation using real host track and weather data for ATL to MIA flights.	May 2010- December 2011	594 trajectories and 56,752 points	Traveling from Hartsfield-Jackson Atlanta International Airport (ATL) to Miami International Airport (MIA).	3D raw trajectories of the flights: source center, date, time, aircraft ID, speed, latitude, longitude, altitude values
[112]	2017	Automatic Surveillance - Broadcast (ADS-B)	ADS-B records over the TMA of Beijing Capital International Airport (BCIA) features three parallel runways	July 2017	36288 flights and 3242384 trajectory points	Beijing Capital International Airport (BCIA)	Type of operation, Beginning time, Aircraft number, Position (X; Y; Z), Heading, Horizontal velocity Vh, Vertical velocity Vv, and more.
[105]	2017	Collected by multiple GDTs (Ground Data Terminals)	GPS and pertinent weather condition data recorded during UAV flight trials	November 1, 2017, to October 28, 2017.	-	Relatively small airspace	Position (X, Y, and Z), Pressure, and Temperature
[72], [191]	2013	FAA Traffic Flow Management System (TFMS)	Contains 4D positions of aircraft - latitude, longitude, altitude, time.	sampling time is 2 minutes	-	George Bush International Airport (IAH) to Logan International Airport (BOS)	Latitude, Longitude, Altitude, Time, Course, Latitude and Longitude Speeds.
[162]	2019	National University of Technology (NTUT) in Taiwan	Assesses user throughput in UAV-assisted communication at 900 MHz using RSS and SNR measurements from UAV transmitters and smartphone receivers	NA	25,023 values	outdoor environment with UAV-BSS on top of three different buildings	RSS and SNR values collected from three different smartphones (Samsung, Huawei, and Apple devices) placed at grid points.
[31]	2021	Automatic Surveillance - Broadcast (ADS-B)	Surveillance information on aerial vehicles	July 10, 2019, from 12:00 PM to 1:00 AM	510,000 items	Liaocheng and Tangshan, China	Trajectory information including longitude, latitude, altitude, ground speed, vertical speed, heading angle, etc
[180]	2022	BlueSky open Air Traffic Simulator (ATS) with a UAV plugin	creation of a UAV-based delivery dataset (UAV-DB) using an Air Traffic Simulator due to the absence of real-world data.	-	7000 deliveries	city of Bogotá, Colombia, where real food deliveries were made, and simulations were performed for distances under 5 km.	Moment of delivery, provider name, expected delivery time, delivery cost, latitude and longitude of starting point, daily traffic, customer location, distance, time to reach the customer, UAV parameters (altitude, ground speed, true airspeed, calibrated airspeed, latitude, longitude, flown distance), and course direction.

usage, making them suitable for real-time applications. Deep learning methods, including RNN [26] and RLSTM [28], require substantial initial computational resources but benefit from the advanced processing power of onboard hardware such as NVIDIA Jetson TX2, ensuring real-time prediction

accuracy without compromising battery life. Other models like LSTM auto-encoders and conflict detection offer moderate computational demands, making them lightweight and energy-efficient for onboard processing. Network formation optimization and trajectory optimization methods balance

TABLE XIII
COMPARISON OF METHODS FOR UAV TRAJECTORY PREDICTION

Model	Accuracy	Prediction Capability	Application Area	Computational Efficiency	Cost	Limitations
Mathematical Models [88], [82], [21], [122], [65], [84], [70], [123], [87], [120]	Moderate	Limited to simple scenarios with known dynamics	Navigation, Control	High	Low	Sensitivity to model assumptions, limited adaptability
Machine Learning Models [145], [142], [139], [68], [99], [29], [150], [153], [179], [180]	Moderate to High	Flexible, data-driven	Various, including surveillance, delivery	Moderate to High	Moderate	Data dependency, generalization challenges (e.g., overfitting)
Deep Learning Models [27], [118], [119], [112], [39], [69], [38], [165], [166], [94], [99], [167], [172], [173], [175], [108], [177], [190], [120], [109]	High	Enhanced feature learning	Image-based, complex environments	Low to Moderate	Moderate to High	Large data requirements, computationally intensive (e.g., GPU usage)
Reinforcement Learning Models [40], [41], [196], [97], [198], [41], [64], [86], [83], [89], [201], [75], [101], [73], [95], [102], [74], [32], [199], [200], [61]	High	Adaptive, learning-based	Autonomous navigation, decision-making	Moderate to High	Moderate to High	Exploration challenges, training complexity (e.g., hyperparameter tuning)

energy usage and processing efficiency through heuristic adjustments and efficient use of onboard resources.

Deploying these methods onboard UAVs requires selecting the appropriate hardware to meet computational demands. Hardware like the NVIDIA Jetson TX2 or Xavier provides the necessary GPU power for running complex algorithms efficiently. In a real-world scenario, the deployment process begins with installing these high-performance embedded systems on the UAVs, then loading the pre-trained models onto the hardware. During operation, the onboard system continuously collects trajectory data from sensors and ADS-B information, which the algorithms process in real-time. For instance, the RLSTM method would use the Jetson's GPU to handle the heavy lifting of recurrent computations, ensuring timely and accurate trajectory predictions. Power management is crucial; hence, these systems are designed to optimize power consumption, balancing processing needs with battery life. Regular updates and model retraining can be managed through periodic data offloading to ground stations for further processing, ensuring the onboard systems remain efficient and up-to-date. This integrated approach, from hardware installation to real-time computation and power management, ensures that UAVs operate safely and effectively in dynamic environments. Utilizing heuristic adjustments and efficient energy usage strategies ensures that UAVs can perform complex trajectory predictions and real-time adjustments despite limited onboard processing capabilities. This leads to improved operational efficiency, longer flight durations, and enhanced safety in dynamic and potentially crowded airspaces.

X. FUTURE DIRECTIONS

The future of UAV trajectory prediction and forecasting will significantly transform UAV's capabilities [28], [72], [110], [174], [178]. Research efforts from various angles converge to usher in a new era of efficiency, precision, and adaptability in UAV operations [29], [56], [57].

One prominent area of exploration involves the coordination of multiple UAVs [2], [157]. By developing advanced strategies for multi-UAV coordination, researchers aim to optimize system efficiency, enabling UAVs to collaborate seamlessly on complex missions during the Mid-flight as most of the accidents occur during this period [25], as shown in Fig. 16. This coordination extends to joint optimization efforts, where trajectories and radio resources for multiple UAVs are fine-tuned to ensure efficient communication and navigation in swarming scenarios. Researchers are delving into integrating higher spatial resolution weather data to enhance trajectory prediction accuracy [239]. This data provides valuable insights into weather patterns and contributes to improved trajectory planning, ensuring UAVs can navigate safely and effectively through changing weather conditions.

The future trajectory prediction landscape also encompasses adaptability and autonomy. Hybrid optimization techniques are under investigation, allowing UAVs to adapt to varying mission parameters and environmental conditions in real-time. Moreover, adaptive energy-efficient trajectory strategies are being developed, considering real-time factors such as cloud coverage, optimizing flight paths, and conserving energy resources. Researchers are exploring integrating broader environment awareness and complex dynamics into trajectory prediction models to enhance real-world navigation. This includes incorporating multi-modal data, such as images, audio, and videos, into deep learning models, thereby increasing prediction accuracy and adaptability across diverse environments [99].

Reinforcement learning is crucial in improving flight state recognition, making UAVs more adaptable to dynamic and unpredictable environments [73], [195]. These developments represent a significant step forward in enhancing UAVs' autonomy and operational capabilities.

The rationale behind prioritizing multi-UAV interactions lies in the potential to address the vulnerabilities identified in mid-flight operations, where accidents are more prevalent [25]. By optimizing collaborative efforts during this critical phase,

TABLE XIV
COMPARATIVE ANALYSIS OF COMPUTATIONAL EFFICIENCY IN UAV TRAJECTORY PREDICTION METHODS

Study	Method	Algorithmic Complexity	Processing Time	Resource Consumption	Similar Studies
[26]	RNN-Based Trajectory Prediction	$O(T \times d \times s)$	Fast for short sequences, increases with sequence length.	Moderate to High due to maintaining and updating RNN states in real-time.	[27], [120], [31], [121], [114]
[102]	TD3-Based UAV Trajectory Design	Training: $O(n^2)$, Execution: $O(n \log n)$	High initially, efficient	High during training: Significant energy and buffer size requirements, Moderate during execution: Efficient energy use and manageable buffer size.	
[28]	RLSTM	Training: $O(T \cdot (d^2 + d \cdot h + h^2))$ Prediction: $O(T \cdot (d \cdot h + h^2))$	Dependent on the specific onboard hardware capabilities (such as an embedded GPU), Number of UAVs being tracked.	Require power-efficient hardware solutions (e.g., NVIDIA Jetson TX2, Xavier) to ensure that the UAV's battery life is not significantly compromised.	[108], [103]
[89]	Continuous Actor-Critic Deep Q-Learning (ACDQL)	NP-hard (Mixed integer non-linear programming problem)	Faster convergence	Lower energy consumption and reduced computational load due to faster convergence and efficient policy updates.	[40], [199], [200]
[105]	LSTM Encoder	$O(n)$	Moderate	Require lightweight and energy-efficient hardware, that provide a balance between performance and power consumption.	[156], [72], [175]
[130]	a. Short-term Trajectory Prediction b. Long-term Trajectory Prediction	a. $O(A \times c(K) \times \log_2(1/\epsilon))$ b. $O(A \times c(K) \times \log_2(1/\epsilon))$	a. High due to frequent updates. b. Lower compared to short-term.	Moderate energy due to continuous processing, High data usage due to frequent updates.	
[41]	LSTM-Based Conflict Detection	$O(T)$	Fast for short sequences	Moderate to High	[98], [108]
[239]	Network Formation Optimization (EDANF)	$O(n \log n)$	Efficient but frequent	Moderate: Buffer size and energy managed by heuristic adjustments.	
[239]	Trajectory Optimization (DRL with Bayesian Optimization)	$O(n^2)$	High initially, efficient post-training	High during training, moderate during execution.	[102], [74], [32], [41]
[73]	Proximal Policy Optimization (PPO)	$O(T)$	Moderate (Depends on episode length)	Moderate to High	[101], [73], [33], [100]
[197]	Double (DDQN)	$O(S \times A \times N)$	1.63612 - 3.95510 seconds per episode	Moderate, efficient for real-time applications, balance between performance and resource usage.	[40], [199], [200]

researchers aim to mitigate risks and improve overall mission success rates. Furthermore, the exploration of reinforcement learning is motivated by its pivotal role in enhancing flight state recognition, enabling UAVs to adapt seamlessly to dynamic and unpredictable environments [73], [195]. The interrelationship between multi-UAV coordination and reinforcement learning is crucial, as coordinated efforts require intelligent decision-making, and reinforcement learning algorithms can contribute to real-time adaptability. Cooperative optimization is envisioned as a key component, ensuring that trajectories and resources are fine-tuned collaboratively to achieve efficient communication and navigation in swarming scenarios. This strategic focus on multi-UAV interactions, reinforcement learning, and cooperative optimization aligns with the broader trajectory prediction landscape, promising a holistic approach to advancing UAV autonomy and operational capabilities. The ongoing trends in this area suggest a promising future for UAVs, not only in trajectory prediction but also in diverse applications spanning aerial surveillance, logistics, and beyond.

One prominent concern in UAV based communication is the potential for cyber-attacks that can compromise the navigation and control systems of UAVs [12], [240]. Research and analysis can be done on robust security protocols and encryption methods for encrypting communications between UAVs and ground control stations, hardening operating systems and software, and using multiple positioning sensors to counter GPS spoofing [241], [242], [243] and jamming [244], [245].

Also, a promising area for future research involves the application of generative AI techniques [246], such as diffusion models, to UAV trajectory prediction. Generative models in UAV trajectory planning enhance prediction accuracy, handle uncertainty, and adapt to dynamic environments by generating multiple plausible and realistic future paths. GAI models like Conditional Variational Autoencoders (CVAEs) can be used in UAV trajectory prediction by capturing the uncertainty in trajectories through latent variables to generate diverse and feasible paths. GANs can enhance UAV trajectory prediction by training a generator to create realistic future trajectories and a discriminator to ensure the accuracy of these predictions.

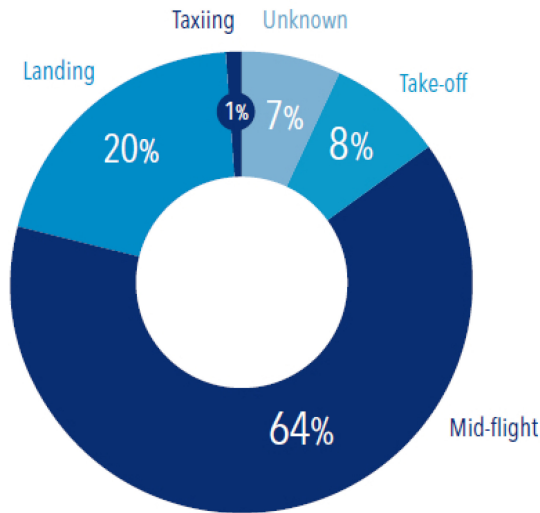


Fig. 16. UAV crash analysis by Drone Wars UK [25].

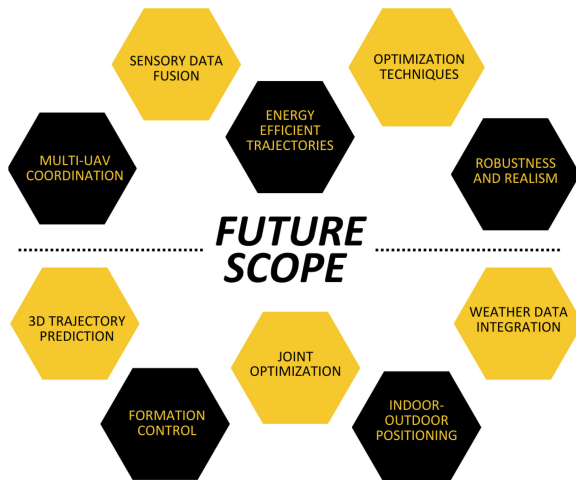


Fig. 17. Future Directions for UAVs Trajectory Prediction.

Transformer-based models can be applied to UAV trajectory prediction to capture long-range dependencies and interactions between multiple agents, making them ideal for modeling temporal sequences. Bivariate Gaussian (BG)-based models offer a computationally efficient method for predicting UAV trajectories by assuming future positions follow a bivariate Gaussian distribution, which is suitable for real-time applications. Hybrid models, such as integrating diffusion models with GANs or CVAEs or any other, can incorporate the strengths of each approach to generate diverse, realistic trajectories and improve prediction accuracy and computational efficiency. When implemented onboard, these models enable real-time trajectory refinement and adaptation to dynamic environments, ensuring safer and more accurate flight paths.

In conclusion, the future of UAV trajectory prediction and forecasting is characterized by a convergence of efforts to optimize coordination, accuracy, adaptability, and autonomy. It depends on many factors as shown in Fig. 17.

These advancements promise to unlock new possibilities for UAVs in various domains, from aerial surveillance to logistics and beyond.

XI. CONCLUSION

This study presents a comprehensive exploration of the trajectory-prediction realm for UAVs and the corresponding taxonomy for trajectory-prediction techniques. The paper delves into an extensive review of trajectory-prediction methods tailored for UAVs, encompassing mathematical models, machine learning-based approaches, deep learning techniques, and reinforcement learning strategies. The efficacy of each method is discussed, and their potential applicability within real-world UAV navigation scenarios is mentioned. Revisiting the original motivations and objectives of this study, our aim was to contribute to the ongoing evolution of UAV trajectory prediction, recognizing its promising advancements while acknowledging the challenges and unexplored research avenues.

Safety remains a paramount concern in the realm of UAV operations. Just as autonomous vehicles need to anticipate their environment like human drivers, UAVs require predictive capabilities to ensure secure and reliable flight. This survey endeavors to enhance the integration of prediction systems in UAVs and spur continued research endeavors, as delineated in the outlined directions. The insights presented here aim to foster the advancement of trajectory prediction systems for UAVs and encourage continued innovation in this evolving domain. By explicitly revisiting the outlined directions for future investigation, we aim to provide a roadmap that can guide researchers and practitioners in navigating the complexities of this domain. The insights presented here are intended to not only enhance the integration of prediction systems in UAVs but also to catalyze continued research endeavors, fostering innovation and progress in this dynamic and evolving field.

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CONFLICT OF INTEREST STATEMENT

The authors of this research affirm that they have no conflicts of interest pertaining to this study. None of the authors have financial interests or affiliations with any organization or entity that could bias or compromise the findings, analysis, or interpretations presented in this manuscript.

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